$$dx^2 + k^2 = p^n, \tag{1}$$

$$\frac{y^n - 1}{y - 1} = x^2 \tag{2.2}$$

$$ds^2 + k^2 = p^r. \tag{A.3}$$

$$db^2 + k^2 = p^{n/q}. \tag{4a}$$

$$4p^{n/q} = a^2 + b^2 d. \tag{5}$$

$$8k = a(a^2 - 3b^2 d). \tag{6.6}$$

$$(y, t, z) \mapsto (t, z). \tag{A.7}$$

$$f_b(0) = D_b^+ + \gamma \text{ and } f_b(\infty) = D_b^- + \gamma, \tag{8a}$$

$$f_b : \mathbb{P}_b^1 \to C^1_{d+u,u,d+u,u}(X_b) \tag{9}$$

$$f_b : \mathbb{P}_b^1 \to C^1_{d+u,u}(X_b) \tag{3.10}$$

$$\widetilde{s} : B \times C^1_{d,d,u,u}(X) \to B \times C^1_{d+u,d+u,u,u}(X) \tag{A.11}$$

$$f_b(0) = (A + C, C) \text{ and } (f_b(\infty)) = (B + D, D). \tag{12a}$$

$$\mathscr{Z}_n = \{(D, b) \mid n[D] = 0\}. \tag{13}$$

$$nz = \sum_i P_{ib} \, . \tag{7.14}$$

$$n[z] = \sum_i n_i [P_{ib}] \tag{A.15}$$

$$t \mapsto t, z \mapsto z \tag{16a}$$

$$\mathbb{Q}[y, t, z]/(y^2 - t^2 q^2 + z^n) \otimes_{\mathbb{Q}[t,z]} \mathbb{Q} \tag{17}$$

$$\mathbb{Z}[y](\sqrt{m^2 q^2 - \ell^n}) \, . \tag{4.18}$$

$$\mathbb{P}^2_{\mathbb{Z}} \times C^1_{d,d}(\widetilde{\bar{S}}_{\mathbb{Z},b}) \tag{A.19}$$

$$f : \mathbb{P}^1_{\mathbb{Z}} \to C^1_{d+u,u,d+u,u}(\widetilde{\bar{S}}_{\mathbb{Z},b}) \tag{20a}$$

$$f : \mathbb{P}^1_{\mathbb{Z}} \to C^1_{d+u,u}(\widetilde{\bar{S}}_{\mathbb{Z},b}) \tag{21}$$

$$f(0) = (A + C, C) \text{ and } (f(\infty)) = (B + D, D). \tag{1.22}$$

$$f(0) = A, \quad f(\infty) = B \, . \tag{A.23}$$

$$4k^2 = 3p^r + 1. \tag{24a}$$

$$\begin{cases} 2k - 1 = 1, \\ 2k + 1 = 3p^r, \end{cases} \tag{25}$$

$$\left(\frac{a + b\sqrt{d}}{2} \right)^q \in \mathbb{Z}(\sqrt{d}) \text{ if and only if } q = 3. \tag{5.26}$$

$$k = \pm 1 \pm \sum_{\ell=1}^{\frac{q-1}{2}} \binom{q}{2\ell} b^{2\ell} (-d)^\ell. \tag{A.27}$$

$$p^n = (k^2 + b^2 d)^q. \tag{28a}$$

$$k + m\sqrt{d} = \left(\frac{a + b\sqrt{-d}}{2} \right)^3. \tag{29}$$

$$p^{n/3} = \frac{2k + 1}{3}, \tag{2.30}$$

$$\cdots \to \pi_*(W^+(k))[u]/(u^n) \to \cdots \to \pi_*(W^+(k))[u]/(u^2) \to \pi_*(W^+(k))[u]/(u) \tag{A.31}$$

$$Mf_{k,pu} \simeq W^+(k) \otimes_{\mathbb{S}_p^\wedge} Mf_{\mathbb{F}_p,pu} \simeq W^+(k) \otimes_{\mathbb{S}_p^\wedge} H\mathbb{F}_p \simeq Hk \tag{32a}$$

$$\cdots \to A/I^n \to \cdots \to A/I^2 \to A/I \tag{33}$$

$$\pi_0(Mf_{R,\xi}) \cong \pi_0(W^+(R^\flat)_{h\Omega^3 S^3}) \cong \pi_0(W^+(R^\flat))_{\pi_0(\Omega^3 S^3)} \tag{6.34}$$

$$\pi_0(W^+(R^\flat))_{\pi_0(\Omega^3 S^3)} \cong W(R^\flat)/(1 - (1 - \xi)) \cong R \tag{A.35}$$

$$\cdots \to A/\mathfrak{m}^n \to \cdots \to A/\mathfrak{m}^2 \to A/\mathfrak{m} \tag{36a}$$

$$0 \to K \to W(k)[[u_1, \ldots, u_n]] \to A \to 0 \tag{37}$$

$$\cdots \to \pi_n(T(M)) \to \pi_n(M) \to \pi_n(M^\wedge_{(x)}) \to \pi_{n-1}(T(M)) \to \pi_{n-1}(M) \to \pi_{n-1}(M^\wedge_{(x)}) \to \cdots \tag{3.38}$$

$$0 \to \overset{1}{\lim} T_{n+1}(M)_* \to \pi_n(T(M)) \to \lim T_n(M)_* \to 0 \tag{A.39}$$

$$\sum_{i=1}^{c} \frac{a_i^3}{b_i} \geq c \cdot \frac{A^3}{B}. \tag{40a}$$

$$V_\tau(\mathcal{T}) \geq \frac{\alpha_V^{3n} \cdot \alpha_B^{3n-o(n)} \cdot \alpha_N^{1.5n-o(n)}}{\max_{\beta \in D_\alpha} \beta_N^{1.5n-o(n)}}. \tag{41}$$

$$\omega < 2.37286. \tag{7.42}$$

$$T = \sum_{i=1}^{k_X} \sum_{j=1}^{k_Y} \sum_{k=1}^{k_Z} T_{ijk}. \tag{A.43}$$

$$\Theta\left(\frac{B}{m}\right) \tag{44a}$$

$$T = \sum_{i=1}^{|X|} \sum_{j=1}^{|Y|} \sum_{k=1}^{|Z|} a_{ijk} \cdot x_i y_j z_k, \tag{45}$$

$$T = \left(\sum_{i=1}^{|X|} \alpha_i \cdot x_i\right) \cdot \left(\sum_{j=1}^{|Y|} \beta_j \cdot y_j\right) \cdot \left(\sum_{k=1}^{|Z|} \gamma_k \cdot z_k\right) \tag{4.46}$$

$$\omega := \inf_{q \in \mathbb{N}} \log_q R(\langle q, q, q \rangle). \tag{A.47}$$

$$\sum_{i=1}^{m} (a_i \cdot b_i \cdot c_i)^\tau = r. \tag{48a}$$

$$CW_q := x_0 y_0 z_{q+1} + x_0 z_{q+1} y_0 + x_{q+1} y_0 z_0 + \sum_{i=1}^{q} (x_0 y_i z_i + x_i y_0 z_i + x_i y_i z_0). \tag{49}$$

$$S' = \{(i, j, k) \in S \mid i \neq j \neq k \neq i \text{ and } i, j, k \in R\}. \tag{1.50}$$

$$\frac{|I|}{n} \cdot \frac{|I|-1}{n-1} \cdot \frac{|I|-2}{n-2} > \left(\frac{|I|-2}{n-2}\right)^3 = \left(\frac{\frac{n \log n}{\sqrt{m}} - 2}{n-2}\right)^3 > \frac{1}{2}\left(\frac{\log n}{\sqrt{m}}\right)^3 \tag{A.51}$$

$$\left(1 - \frac{2m}{N^2}\right)^{N^3 (\log^3 N)/(16 m^{3/2})} \leq 2^{-\Omega(\frac{N \log^3 N}{\sqrt{m}})}. \tag{52a}$$

$$\alpha_{X_i} := \sum_{j \in [k_Y], k \in [k_Z] | (i,j,k) \in S} \alpha_{ijk}, \tag{53}$$

$$\alpha_{V_\tau} := \prod_{(i,j,k) \in S} V_\tau (T_{ijk})^{\alpha_{ijk}} \text{ for } \tau \in [2/3, 1]. \tag{5.54}$$

$$\frac{\alpha_B^{3n-o(n)} \cdot \alpha_N^{1.5n-o(n)}}{\max_{\beta \in D_\alpha} \beta_N^{1.5n-o(n)}} \tag{A.55}$$

$$\alpha_{X_i} = \frac{1}{n} \cdot |\{\ell \in [n] \mid I_\ell = i\}|. \tag{56a}$$

$$N_B = \alpha_B^{3n-o(n)}. \tag{57}$$

$$N_\alpha = \alpha_N^{3n-o(n)}. \tag{2.58}$$

$$\mathbb{E}[C_1'] \geq \mathbb{E}[C_1 - 2C_2] = \mathbb{E}[C_1] - 2\mathbb{E}[C_2] \geq \frac{|A| \cdot N_\alpha}{M^2} - 2\frac{3 \cdot |A| \cdot N_\alpha}{200M^2} = \frac{97 \cdot |A| \cdot N_\alpha}{100 \cdot M^2}. \tag{A.59}$$

$$V_\tau(\mathcal{T}'''_{X_B,Y_B,Z_B}) \geq \prod_{(i,j,k)\in S} V_\tau(T_{ijk} \otimes T^r_{ijk} \otimes T^{rr}_{ijk})^{\alpha_{ijk}\cdot n} = \prod_{(i,j,k)\in S} V_\tau(T_{ijk})^{3\alpha_{ijk}\cdot n} = \alpha_V^{3n}. \tag{60a}$$

$$\alpha_{V_\tau} \cdot \alpha_B \cdot \sqrt{\alpha_N} \tag{61}$$

$$\gamma_{V_\tau} \cdot \gamma_B + \lambda/\gamma_N \tag{6.62}$$

$$\sum_{i=1}^{q} x_{0,i}y_{0,i}z_{q+1,0} + \sum_{i=1}^{q} x_{i,0}y_{i,0}z_{0,q+1} + \sum_{i=1}^{q}\sum_{j=1}^{q}(x_{0,j}y_{i,0}z_{i,j} + x_{i,0}y_{0,j}z_{i,j}). \tag{A.63}$$

$$T^2_{112} = T_{002} \otimes T_{110} + T_{110} \otimes T_{002} + T_{011} \otimes T_{101} + T_{101} \otimes T_{011}. \tag{64a}$$

$$T^2_{220} \equiv x_{0,0}y_{0,0}z_{0,0} + x_{q+1,q+1}y_{q+1,q+1}z_{0,0} + \sum_{i=1}^{q} x_{i,i}y_{i,i}z_{0,0} = \sum_{i=0}^{q+1} x_{i,i}y_{i,i}z_{0,0} \equiv \langle q+2, 1, 1\rangle. \tag{65}$$

$$\frac{1}{2^{o(n)}} \leq \frac{\alpha_N}{\alpha'_N} \leq 2^{o(n)}. \tag{3.66}$$

$$\frac{1}{2^{o(n)}} \leq \frac{\alpha_{ijk}^{-\alpha_{ijk}}}{{\alpha'}_{ijk}^{-\alpha'_{ijk}}} \leq 2^{o(n)}. \tag{A.67}$$

$$\frac{\alpha_{ijk}^{-\alpha_{ijk}}}{{\alpha'}_{ijk}^{-\alpha'_{ijk}}} \leq 2^{O(1/n)}. \tag{68a}$$

$$\left(\frac{1}{c}\sum_{i=1}^{c} a_i^{3/2}\right)^{2/3} \geq \frac{1}{c}\sum_{i=1}^{c} a_i = A, \tag{69}$$

$$x^* \in C, y^* \in Q \ \text{ such that } \ Ax^* = By^*. \tag{7.70}$$

$$x_{n+1} = x_n - \gamma_n[(I - P_C) + A^*(I - P_Q)A)]x_n, \ \forall n \geq 0, \tag{A.71}$$

$$\begin{cases} x_{n+1} = P_C(x_n - \gamma_n A^*(Ax_n - By_n)), \\ y_{n+1} = P_Q(y_n + \gamma_n B^*(Ax_{n+1} - By_n)), \end{cases} \tag{72a}$$

$$\begin{cases} x = x - \gamma\left((I - P_C)x + A^*(Ax - By)\right), \\ y = y - \gamma\left((I - P_Q)y - B^*(Ax - By)\right). \end{cases} \tag{73}$$

$$\theta_{n+1} \leq \delta_n\left(\|u_n - x^*\|^2 + \|v_n - y^*\|^2\right) + (1 - \delta_n)\theta_n \leq \max\{G, \theta_n\} \leq \cdots \leq \max\{G, \theta_0\}. \tag{4.74}$$

$$\lim_{k\to\infty}\|(I - P_C)x_{n_k}\| = \lim_{k\to\infty}\|(I - P_Q)y_{n_k}\| = \lim_{k\to\infty}\|Ax_{n_k} - By_{n_k}\| = 0. \tag{A.75}$$

$$\limsup_{k\to\infty}\langle v_{n_k} - y^*, y_{n_k} - y^*\rangle = \lim_{j\to\infty}\langle v_{n_{k_j}} - y^*, y_{n_{k_j}} - y^*\rangle = \langle v - y^*, \bar{y} - y^*\rangle \leq 0. \tag{76a}$$

$$\|y_{n_k+1}-y_{n_k}\| \le \delta_{n_k}\|v_{n_k}-y_{n_k}\| + (1-\delta_{n_k})\|\widehat{y}_{n_k}-y_{n_k}\| \to 0. \tag{77}$$

$$\|\widehat{x}_n - x^*\|^2 + \|\widehat{y}_n - y^*\|^2 = \|x_n - x^*\|^2 + \|y_n - y^*\|^2 - \Phi_n \le \|x_n - x^*\|^2 + \|y_n - y^*\|^2. \tag{1.78}$$

$$\|y_{n_k+1}-y_{n_k}\| = \delta_{n_k}\|g(y_{n_k})-y_{n_k}\| + (1-\delta_{n_k})\|\widehat{y}_{n_k}-y_{n_k}\| \to 0, n \to \infty. \tag{A.79}$$

$$\begin{cases} x_{n+1} = P_{C_n}(x_n - \gamma A^*(Ax_n - By_n)), \\ y_{n+1} = P_{Q_n}(y_n + \gamma B^*(Ax_{n+1} - By_n)), \end{cases} \tag{80a}$$

$$\|f(x) - f(y)\| \le \lambda\|x - y\|,\ \forall x, y \in H,\ \lambda \in [0, 1); \tag{81}$$

$$\|x + y\|^2 \le \|x\|^2 + 2\langle y, x + y\rangle; \tag{5.82}$$

$$\begin{cases} 0 = (I - P_C)x + A^*(Ax - By), \\ 0 = (I - P_Q)y - B^*(Ax - By). \end{cases} \tag{A.83}$$

$$0 = \langle (I - P_Q)y - B^*(Ax - By), y - y^*\rangle \ge \|y - P_Q y\|^2 - \langle Ax - By, By - By^*\rangle. \tag{84a}$$

$$\theta_{n+1} \le \theta_n - \eta_n + \zeta_n, n \ge 0, \tag{85}$$

$$2\langle Ax_n - By_n, By_n - By^*\rangle - 2\langle Ax_n - By_n, Ax_n - Ax^*\rangle = -2\|Ax_n - By_n\|^2. \tag{2.86}$$

$$\begin{aligned} \Phi_n = {} & 2\gamma_n(1-\gamma_n)\left(\|(I-P_C)x_n\|^2 + \|(I-P_Q)y_n\|^2\right) + 2\gamma_n\|Ax_n - By_n\|^2 \\ & - 2\gamma_n^2\left(\|A^*(Ax_n - By_n)\|^2 + \|B^*(Ax_n - By_n)\|^2\right). \end{aligned} \tag{A.87}$$

$$\|y_{n+1} - y^*\|^2 \le (1-\delta_n)\|\widehat{y}_n - y^*\|^2 + 2\delta_n\langle v_n - y^*, y_{n+1} - y^*\rangle. \tag{88a}$$

$$\limsup_{k\to\infty} \langle u_{n_k} - x^*, x_{n_k} - x^*\rangle = \lim_{j\to\infty}\langle u_{n_{k_j}} - x^*, x_{n_{k_j}} - x^*\rangle, \tag{89}$$

$$\|\widehat{x}_{n_k} - x_{n_k}\| \le \gamma_n(\|(I-P_C)x_{n_k}\| + \|A\|\|Ax_{n_k} - By_{n_k}\|) \to 0, \tag{6.90}$$

$$\|\widehat{y}_{n_k} - y_{n_k}\| \le \gamma_n(\|(I-P_Q)y_{n_k}\| + \|B\|\|Ax_{n_k} - By_{n_k}\|) \to 0. \tag{A.91}$$

$$Q_n = \{y \in H_2 | \mathfrak{g}(y_n) + \langle \eta_n, y - y_n\rangle \le 0\},\ \eta_n \in \partial\mathfrak{g}(y_n). \tag{92a}$$

$$\begin{aligned} \|x_{n+1} - x^*\|^2 + \|y_{n+1} - y^*\|^2 \le {} & (1-\delta_n)\left(\|x_n - x^*\|^2 + \|y_n - y^*\|^2\right) - (1-\delta_n)\Psi_n \\ & + 2\delta_n\left(\langle u_n - x^*, x_{n+1} - x^*\rangle + \langle v_n - y^*, y_{n+1} - y^*\rangle\right). \end{aligned} \tag{93}$$

$$\theta_{n+1} \le (1-\delta_n)\theta_n + \delta_n\tau_n, n \ge 0 \text{ and } \theta_{n+1} \le \theta_n - \eta_n + \zeta_n, n \ge 0. \tag{3.94}$$

$$\limsup_{k\to\infty} \langle g(y^*) - y^*, y_{n_k} - y^*\rangle = \lim_{j\to\infty}\langle g(y^*) - y^*, y_{n_{k_j}} - y^*\rangle. \tag{A.95}$$

$$\limsup_{k\to\infty} \langle g(y^*) - y^*, y_{n_k} - y^*\rangle = \lim_{j\to\infty}\langle g(y^*) - y^*, y_{n_{k_j}} - y^*\rangle = \langle g(y^*) - y^*, \bar{y} - y^*\rangle \le 0. \tag{96a}$$

$$\|\widehat{y}_{n_k} - y_{n_k}\| \le \gamma_n(\|(I-P_Q)y_{n_k}\| + \|B\|\|Ax_{n_k} - By_{n_k}\|) \to 0, n \to \infty. \tag{97}$$

$$\gamma_n = 0.65 \times \min\left\{ \frac{\|Ax_n - By_n\|^2}{\|A^*(Ax_n - By_n)\|^2}, \frac{\|Ax_n - By_n\|^2}{\|B^*(Ax_n - By_n)\|^2} \right\}. \tag{7.98}$$

$$\delta_n = \frac{1}{n+50},\ f(x) = 0.6x,\ g(y) = 0.6y. \tag{A.99}$$

$$N(G)\mathbf{x}_S = \mathbf{x}_C \tag{100a}$$

$$pr(A) = \sum_{i,j} A_{ij} = \sum_i A_{ii} = \sum_i 1 = pr(n). \tag{101}$$

$$pr(A^{-1}) = \sum_i (A^{-1})_{ii}. \tag{4.102}$$

$$\nu(G_1 \oplus G_2) = \nu(G_1) + \nu(G_2) + pr(\rho(G_1)\rho(G_2)). \tag{A.103}$$

$$\rho(G_1 \oplus G_2) = \rho(G_1) + \rho(G_2) - pr(\rho(G_1)\rho(G_2)). \tag{104a}$$

$$\begin{aligned} pr(n) = Tr(I) &= Tr(AA^{-1}) \\ &= \sum_{i,j} A_{ij}(A^{-1})_{ij} \\ &= \sum_i A_{ii}(A^{-1})_{ii} \\ &= \sum_i (A^{-1})_{ii}. \end{aligned} \tag{105}$$

$$\nu(G_1 \oplus ... \oplus G_m) = \left\{ \begin{array}{ll} \sum_{i=1}^m \nu(G_i) & \text{if} \quad j = 0 \\ \sum_{i=1}^m \nu(G_i) + j - 1 & \text{otherwise} \end{array} \right\}. \tag{1.106}$$

$$\rho(G_1 \oplus ... \oplus G_m) = \left\{ \begin{array}{ll} \sum_{i=1}^m \rho(G_i) & \text{if} \quad j = 0 \\ \sum_{i=1}^m \rho(G_i) - j + 1 & \text{otherwise} \end{array} \right\}. \tag{A.107}$$

$$N[u] = \{v \in V(G) \,|v \text{ is adjacent to } u \text{ or } v = u\}. \tag{108a}$$

$$pr(S) = pr(\mathbf{x}_S) = pr(N^{-1}\mathbf{1}) = pr(N^{-1}) = pr(N) = pr(n) = pr(\rho(G)). \tag{109}$$

$$\begin{aligned} \overline{NCD}_i^{t\omega} &\geq NCD_i^{t\omega} \\ \overline{NCD}_i^{t\omega} &\geq 0. \end{aligned} \tag{5.110}$$

$$H_i^{t\omega} = \max\{0, S_i^t + R_i^t - D_i^{(t-1)} - d_i^{t\omega}\}, \quad \text{for all } \omega \in \Omega, i \in N, t \in T, \tag{A.111}$$

$$NCD_i^{t\omega} = d_i^{t\omega} + D_i^t - S_i^t - R_i^t, \quad \text{for all } \omega \in \Omega, i \in N, t \in T. \tag{112a}$$

$$\overline{NCD}_i^{t\omega} = \max\{0, NCD_i^{t\omega}\}, \quad \text{for all } \omega \in \Omega, i \in N, t \in T. \tag{113}$$

$$\Phi_1(s, x, y; \Omega) = \max_{i \in N} \sum_{\omega \in \Omega} p^\omega \sum_{t \in T} \overline{NCD}_i^{t\omega} \tag{2.114}$$

$$\Phi_2(s,x,y;\Omega) = \max_{i\in N}\max_{t\in T}\sum_{\omega\in\Omega} p^{\omega}\overline{NCD}_i^{t\omega} \tag{A.115}$$

$$\Phi_3(s,x,y;\Omega) = \max_{k=1,\dots,L}\sum_{\omega\in\Omega} p^{\omega}\sum_{i\in M_k}\sum_{t\in T}\overline{NCD}_i^{t\omega} \tag{116a}$$

$$\Phi_4(s,x,y;\Omega) = \sum_{\omega\in\Omega} p^{\omega}\sum_{i\in N}\sum_{t\in T}\overline{NCD}_i^{t\omega} \tag{117}$$

$$H_i^{t\omega} = \max\{0, S_i^t + R_i^t - D_i^{(t-1)} - d_i^{t\omega}\}, \tag{6.118}$$

$$h_i^t(k) = S_i^t(k) + R_i^t(k) - D_i^{(t-1)}(k), \ \ \forall i\in N, \omega\in\Omega, \tag{A.119}$$

$$I_s(\mu) = \iint \frac{d\mu(x)\, d\mu(y)}{|x-y|^s}. \tag{120a}$$

$$P_a(x) := \frac{x-a}{|x-a|}. \tag{121}$$

$$\psi\{\alpha : |\widetilde{\pi_\alpha}(x) - \widetilde{\pi_\alpha}(y)| \le \delta|x-y|\} < c\cdot\delta^m\cdot|x-y|^{m-s}, \tag{3.122}$$

$$\psi(T_\delta) \lesssim \delta^{n-1}. \tag{A.123}$$

$$\Phi_\lambda(\mathbf{a}) = a_2 + \gamma(\lambda - a_1). \tag{124a}$$

$$d_\lambda := \operatorname{dist}(\mathbf{x}, \ell_\lambda) = |\lambda - x_1|, \tag{125}$$

$$\Phi_\lambda(\mathbf{b}) - \Phi_\lambda(\mathbf{a}) > 0 \tag{7.126}$$

$$p = \frac{b_1 - a_1}{2}, \quad \text{and} \quad q = s_0. \tag{A.127}$$

$$\Phi_\lambda(\mathbf{b}) = b_2 + \gamma(\lambda - b_1) = b_2 - d + \gamma(\lambda - b_1) + d = \Phi_\lambda((b_1, b_2 - d)) + d. \tag{128a}$$

$$\Phi_\alpha(\mathbf{a}) = a_3 + \gamma(\alpha_1 - a_1, \alpha_2 - a_2). \tag{129}$$

$$\int_X D(\widetilde{\pi_\alpha}_\sharp\mu, h, u)^2\, dh(u) = \int_X D(\widetilde{\pi_\alpha}_\sharp\mu, h, u)\, d\widetilde{\pi_\alpha}_\sharp\mu(u). \tag{4.130}$$

$$\begin{aligned}\Phi_\lambda(\mathbf{a}) - \Phi_\lambda(\mathbf{b}) = &\sum_{j=0}^{1}\left(\gamma'(\lambda - a_1 - jp - h_j p)\cdot p\right)\\ &-\sum_{j=0}^{1}\left(\gamma'(q - jp - h'_j p)\cdot p\right),\end{aligned} \tag{A.131}$$

$$1 \le h(\widetilde{\pi_\alpha}F)\int_{\widetilde{\pi_\alpha}F} D(\widetilde{\pi_\alpha}_\sharp\mu, h, u)^2\, dh(u) \tag{132a}$$

$$\operatorname{Fav}(F(r)) \gtrsim r^{1-s} \tag{133}$$

$$\mathrm{Fav}_{\Gamma,d}(e) \sim |E+\Gamma|_d. \tag{1.134}$$

$$\mu(B(x,r)) \lesssim r^t \tag{A.135}$$

$$\int_A \mathrm{vis}(a, E(r))\, d\mathcal{H}^{n-1}(a) \gtrsim (\log r^{-1})^{-1}. \tag{136a}$$

$$\mathrm{Fav}_\Gamma(\mathcal{K}_n) \gtrsim \frac{1}{n}. \tag{137}$$

$$\frac{\sin\theta}{|x-y|} = \frac{\sin\gamma}{|a-x|} \tag{5.138}$$

$$|P_a(x) - P_a(y)| \geq c\theta \geq c \cdot \frac{\rho}{4R^2} \cdot |x-y|. \tag{A.139}$$

$$d_\lambda \cdot |\mathbf{a}-\mathbf{b}| \lesssim \delta, \tag{140a}$$

$$\eta - \xi < (\lambda - a_1) - t_0 = (\lambda - a_1 - s_0) + (s_0 - t_0) = d_\lambda + (b_1 - a_1). \tag{141}$$

$$(\lambda - a_1 - jp - h_j p) - (s_0 - jp - h'_j p) = d_\lambda - h_j p + h'_j p \leq 3d_\lambda/2. \tag{2.142}$$

$$(\lambda - a_1 - jp - h_j p) - (q - jp - h'_j p) = b_1 - a_1 - h_j p + h'_j p \geq (b_1 - a_1)/2, \tag{A.143}$$

$$d := h(\widehat{\lambda}). \tag{144a}$$

$$|\{\lambda \in A : h(\lambda) \leq \delta\}| \leq |A| \leq c \leq c\frac{c'\,\delta}{|\mathbf{b}-\mathbf{a}|}, \tag{145}$$

$$\gamma_{\lambda_2} := \Gamma \cap \{y = \lambda_2\} = \{(t_1, \lambda_2, \gamma(t_1, \lambda_2)) : (t_1, \lambda_2) \in B\} \tag{6.146}$$

$$\int_A I_\tau(\widetilde{\pi_\alpha}_\sharp \mu)\, d\alpha \lesssim \iint \frac{d\mu(x)\, d\mu(y)}{|x-y|^\tau} = I_\tau(\mu) < \infty, \tag{A.147}$$

$$\int g\, df_\sharp \nu = \int (g \circ f)\, d\nu \tag{148a}$$

$$\mu(B(x,r)) \leq cr^s \tag{149}$$

$$A\{y\} \longrightarrow A\{y\} \otimes^{\mathbb{L}}_{B_0} C_0 \longrightarrow A\{y\} \otimes^{\mathbb{L}}_{B_1} C_1 \longrightarrow \cdots \tag{3.150}$$

$$B_n = A_n[u] \to A_n[u,v]/(u - \varphi^n(d)v - P_n(z, \delta(z), \ldots, \delta^{n-1}(z))) \cong A_n[v] = C_n \tag{A.151}$$

$$L_{B/A} \otimes^{\mathbb{L}}_B R \longrightarrow L_{R/A} \longrightarrow L_{R/B} \tag{152a}$$

$$(\Gamma_{\mathbb{Z}}(T) \twoheadrightarrow \mathbb{Z}, \gamma) \to (\Gamma_{\mathbb{Z}}(T_0, \ldots, T_n) \twoheadrightarrow \mathbb{Z}, \gamma) \to (\Gamma_{\mathbb{Z}}(X,Y) \twoheadrightarrow \mathbb{Z}, \gamma) \tag{153}$$

$$(\mathbb{Z}/4\mathbb{Z})[1] \oplus \mathbb{Z}/4\mathbb{Z} \twoheadrightarrow \mathbb{F}_2[1] \oplus \mathbb{F}_2 \tag{7.154}$$

$$\Gamma_{\mathbb{Z}[X]}(Y) \otimes^{\mathbb{L}}_{\mathbb{Z}} \Gamma_{\mathbb{Z}[X']}(Y') \simeq \Gamma_{\mathbb{Z}[X,X']}(Y, Y') \tag{A.155}$$

$$\mathbb{Z}[X] \otimes^{\mathbb{L}}_{\mathbb{Z}} \mathbb{Z}[X'] \simeq \mathbb{Z}[X, X'] \tag{156a}$$

$$(M \otimes^{\mathbb{L}}_{A} N)/^{\mathbb{L}}p \simeq (M/^{\mathbb{L}}p) \otimes^{\mathbb{L}}_{A/^{\mathbb{L}}p} (N/^{\mathbb{L}}p) \tag{157}$$

$$\pi_i((M/^{\mathbb{L}}p) \otimes^{\mathbb{L}}_{A/^{\mathbb{L}}p} (N/^{\mathbb{L}}p)) \simeq \pi_0(M/^{\mathbb{L}}p) \otimes_{\pi_0(A/^{\mathbb{L}}p)} \pi_i(N/^{\mathbb{L}}p) \cong 0 \tag{4.158}$$

$$\theta(\delta_n(x)) = \delta_{n-1}(x)\theta(x) \tag{A.159}$$

$$L_{\varphi^*_A(P^\bullet)/A} \otimes^{\mathbb{L}}_{\varphi^*_A(P^\bullet)} R_1 \longrightarrow L_{R_1/A} \longrightarrow L_{R_1/\varphi^*_A(P^\bullet)} \tag{160a}$$

$$L_{\varphi^*_A(P^\bullet)/A} \otimes^{\mathbb{L}}_{\varphi^*_A(P^\bullet)} M \longrightarrow L_{R_1/A} \otimes^{\mathbb{L}}_{R_1} M \longrightarrow L_{R_1/\varphi^*_A(P^\bullet)} \otimes^{\mathbb{L}}_{R_1} M \tag{161}$$

$$N^\bullet \longrightarrow N^\bullet \otimes^{\mathbb{L}}_{B^\bullet} C^\bullet \tag{1.162}$$

$$L_{P_0/A_0} \otimes^{\mathbb{L}}_{P_0} R_0 \longrightarrow L_{R_0/A_0} \longrightarrow L_{R_0/P_0} \tag{A.163}$$

$$L_{P_0/A_0} \otimes^{\mathbb{L}}_{P_0} M \longrightarrow L_{R_0/A_0} \otimes^{\mathbb{L}}_{R_0} M \longrightarrow L_{R_0/P_0} \otimes^{\mathbb{L}}_{R_0} M \tag{164a}$$

$$\varphi^*_{P_0 \twoheadrightarrow P_0 \otimes^{\mathbb{L}}_{A_0} A''_0}(R_0) \simeq R_1 \otimes^{\mathbb{L}}_{\varphi^*_{A_0}(P_0)} P_0 \tag{165}$$

$$D_{\mathbb{Z}_{(p)}\{x\}}(x) \longrightarrow \mathbb{Z}_{(p)}\{x, \varphi(x)/p\} \tag{5.166}$$

$$\delta^n(y) = \delta^n(z)\varphi^n(d) + P_n(z, \delta(z), \ldots, \delta^{n-1}(z)) \tag{A.167}$$

$$\delta((\delta^{n-2}(z))^p) = \sum_{k=1}^{p} \binom{p}{k} (\delta^{n-2}(z))^{p(p-k)} p^{k-1} (\delta^{n-1}(z))^k \tag{168a}$$

$$L_{(P'' \otimes^{\mathbb{L}}_{B''} C'')/C''} \otimes^{\mathbb{L}}_{P'' \otimes^{\mathbb{L}}_{B''} C''} C'' \to L_{C''/C''} \simeq 0 \to L_{C''/(P'' \otimes^{\mathbb{L}}_{B''} C'')} \tag{169}$$

$$\S AB + \S AC + \S BC \geq \S A + \S B + \S C + \S ABC. \tag{2.170}$$

$$\S ABDG + \S ACEG + \S BCFG \geq \S A + \S B + \S C + \S ABCDEFG. \tag{A.171}$$

$$\sum_{l=1}^{L} \alpha_l S_{I_l} \geq \sum_{r=1}^{R} \beta_r S_{J_r}, \tag{172a}$$

$$\sigma(x) = \bigcap_{l=1}^{L} \sigma_{I_l}^{x_l}, \qquad \sigma_I^b = \begin{cases} \sigma_I & \text{if} \quad b = 1, \\ \sigma_I^{\complement} & \text{if} \quad b = 0. \end{cases} \tag{173}$$

$$\sigma_{I_l} = \bigcup_{x : x_l = 1} \sigma(x). \tag{6.174}$$

$$x_l^{(i)} = \delta(i \in I_l). \tag{A.175}$$

$$\sigma_{J_r} = \bigcup_{x : f(x)_r = 1} \sigma(x), \tag{176a}$$

$$f(x^{(i)})_r = \delta(i \in J_r), \tag{177}$$

$$\gamma_{I_l} = \bigcup_{x,x' : x_l \neq x'_l} \gamma(x, x'). \tag{3.178}$$

$$\sum_{x,x'} d_\alpha(x,x') \mathcal{A}\left[\gamma(x,x')\right] \geq \sum_{x,x'} d_\beta(f(x),f(x')) \mathcal{A}\left[\gamma(x,x')\right]. \tag{A.179}$$

$$d_\alpha(x,x') \geq d_\beta(f(x),f(x')). \tag{180a}$$

$$\S ABG + \S ACG + \S BCG \geq \S A + \S B + \S C + \S ABCG, \tag{181}$$

$$\S ABDG + \S ACG + \S BCG \geq \S A + \S B + \S C + \S ABCDG, \tag{7.182}$$

$$\S ABDG + \S ABCG \geq \S ABG + \S ABCDG. \tag{A.183}$$

$$\S ABDG + \S ACEG + \S BCG \geq \S A + \S B + \S C + \S ABCDEG, \tag{184a}$$

$$\S ACEG + \S ABCDG \geq \S ACG + \S ABCDEG. \tag{185}$$

$$\S BCFG + \S ABCDEG \geq \S BCG + \S ABCDEFG. \tag{4.186}$$

$$Ax \geq 0 \qquad \text{where} \qquad A \in \mathbb{R}^{m \times n}, \quad x \in \mathbb{R}^n_{\geq 0}, \quad 0 \in \mathbb{R}^n. \tag{A.187}$$

$$U = \{x \in \mathbb{R}^n_{\geq 0} \ : \ Ax \geq 0\}. \tag{188a}$$

$$\min\{b^T x \ : \ x \in U\} \geq 0, \tag{189}$$

$$A^T q = b. \tag{1.190}$$

$$\S A + \S B \geq \S AB. \tag{A.191}$$

$$Q(I; J:K) = -I_3(I:J:K) + I_2(J:K). \tag{192a}$$

$$\S AB + \S BC \geq \S B + \S ABC. \tag{193}$$

$$\begin{gathered} \S AG + \S BCG \geq \S A + \S BC, \\ \S ABG + \S ACG + \S BC \geq \S AG + \S B + \S C + \S ABCG, \end{gathered} \tag{5.194}$$

$$\S AB + \S AC + \S AD + \S BC + \S BD \geq \S A + \S B + \S CD + \S ABC + \S ABD. \tag{A.195}$$

$$S_I + S_J \geq S_{I \cup J} \ . \tag{196a}$$

$$S_I + S_J \geq S_{I \cap J} + S_{I \cup J} \ . \tag{197}$$

$$S_{I \cup J} + S_{I \cup K} + S_{J \cup K} \geq S_I + S_J + S_K + S_{I \cup J \cup K} \ . \tag{2.198}$$

$$S = \{S_I \ : \ \varnothing \neq I \subseteq [n]\}, \tag{A.199}$$

$$\begin{aligned}\tilde{S}_1 &= \frac{1}{4}(\S1+\S2+\S3+\S4) = \frac{1}{4}(\S1+\S2+\S3+\S123),\\ \tilde{S}_2 &= \frac{1}{6}(\S12+\S13+\S14+\S23+\S24+\S34) = \frac{1}{3}(\S12+\S13+\S23),\end{aligned} \tag{200a}$$

$$\tilde{G} = \frac{1}{(n+1)!}\bigoplus_{\sigma\in Sym_{n+1}} G^{\sigma}, \tag{201}$$

$$\tilde{q}_k = \sum_{J\in Q_n(k)} q_J. \tag{6.202}$$

$$M = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \end{bmatrix}, \qquad N = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}, \tag{A.203}$$

$$\begin{aligned}\tilde{S}_k &= \binom{n+1}{k}^{-1}\left(\binom{n}{k}S_k + \binom{n}{n+1-k}S_{n+1-k}\right)\\ &= \frac{1}{n+1}\left((n+1-k)\min\{k, n-k+w\} + k\min\{n+1-k, w+k-1\}\right)\ .\end{aligned} \tag{204a}$$

$$\tilde{S}_k^{(l)} = \frac{2k}{n+1}(n+1-\max\{k,l\}). \tag{205}$$

$$2\tilde{S}_1 - \tilde{S}_2 \geq 0\ . \tag{3.206}$$

$$-3\tilde{S}_1 + 3\tilde{S}_2 - \tilde{S}_3 \geq 0\ . \tag{A.207}$$

$$-6\tilde{S}_2 + 8\tilde{S}_3 - 3\tilde{S}_4 \geq 0\ . \tag{208a}$$

$$\begin{aligned} S_1 + S_2 &\geq S_{12}\ ,\\ S_{12} + S_{23} &\geq S_2 + S_{123}\ .\end{aligned} \tag{209}$$

$$\begin{cases} -S_{i-1} + 2S_i - S_{i+1} \geq 0 & \forall\ 1 \leq i \leq n-1 \\ -S_{i-1} - S_{n-i} + S_i + S_{n-i+1} \geq 0 & \forall\ 1 \leq i \leq n-1 \end{cases} \tag{7.210}$$

$$1 \leq a \leq n \qquad\qquad \max(a, n-a) \leq b \leq n\ . \tag{A.211}$$

$$M_{10}^{\mathrm{SQEC}} = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}\ , \tag{212a}$$

$$\tilde{S}_k^{(l)} = \min\{k,l\}\ . \tag{213}$$

$$\Pi(M_n^{\mathrm{SQEC}}) = \frac{(2d_n)!}{2^{d_n}}. \tag{4.214}$$

$$\Pi(M_n^{\mathrm{SHEC}}) = \left(-\frac{1}{6}\right)^{d_n}\prod_{k=1}^{3}(1-x_k)_{d_n}\ , \tag{A.215}$$

$$\prod_{k=1}^{3}(x-x_k)=x^3-x-d_n(d_n+1)(3n-2(d_n-1)). \tag{216a}$$

$$\prod_{k=1}^{3}(\hat{x}-\hat{x}_k)\approx \hat{x}^3-4\ . \tag{217}$$

$$\Pi(M_n^{\rm SHEC})\approx\left(-\frac{1}{6}\right)^{d_n}\prod_{k=1}^{3}(1-\hat{x}_k d_n)_{d_n}\ , \tag{1.218}$$

$$(1-d_n\hat{x}_k)_{d_n}\approx\left(1-\frac{1}{\hat{x}_k}\right)^{\frac{1}{2}+(1-\hat{x}_k)d_n}\left(-\frac{\hat{x}_k d_n}{e}\right)^{d_n}. \tag{A.219}$$

$$\Pi(M_n^{\rm SHEC})\approx\frac{\sqrt{3}}{2}\left(\frac{3}{5}\right)^{d_n}d_n^{3d_n}\sim e^{-\frac{13}{10}n}n^{\frac{3}{2}n}. \tag{220a}$$

$$\begin{aligned}-S_{12}-S_{13}-S_{14}-S_{25}-S_{26}-S_{35}+S_{123}+S_{124}+S_{125}+S_{126}\\ +S_{134}+S_{135}+S_{235}+S_{256}-S_{1234}-S_{1235}-S_{1256}\geq 0\ ,\end{aligned} \tag{221}$$

$$F(t,\vec{x})=\langle V(0)W(t,\vec{x})V(0)W(t,\vec{x})\rangle, \tag{5.222}$$

$$\lambda_L^{\pm}=\frac{2\pi}{\beta}\frac{1}{1\mp\ell\Omega}. \tag{A.223}$$

$$\lambda_L^{-}\leq\frac{2\pi}{\beta}\leq\lambda_L^{+}. \tag{224a}$$

$$OTOC(t,\varphi_{12})\approx 1+C_1e^{\frac{2\pi}{\beta_+}(t+\ell\varphi_{12})}+C_2e^{\frac{2\pi}{\beta_-}(t-\ell\varphi_{12})}, \tag{225}$$

$$g_{WAdS}=g_{AdS_3}-2H^2\xi\otimes\xi, \tag{2.226}$$

$$x^+=e^{\frac{2\pi}{\beta}w^+},\qquad x^-=w^-+\left(\frac{\bar{\beta}}{\beta}-\frac{2\pi\mu}{\beta}\right)w^+. \tag{A.227}$$

$$-iP_0^{\rm vac}\left(\frac{\bar{\beta}}{\beta}-\frac{2\pi\mu}{\beta}\right)=\mu q_\psi,\qquad \beta=\tilde{\beta}_\psi\equiv\frac{1\pi}{\sqrt{24\tilde{h}_\psi/c-1}}. \tag{228a}$$

$$\lim\left|\frac{d}{ds}\log F_{ij}(s)\right|\leq 2\pi, \tag{229}$$

$$\mathcal{L}_{\zeta_\pm}g_{\mu\nu}\Big|_{\rm RT}\approx 0. \tag{6.230}$$

$$\mathbf{S}(\rho)=\{S_{\mathscr{I}}\text{ for all }\mathscr{I}\}. \tag{A.231}$$

$$S_{12}+S_{23}\geq S_2+S_{123}. \tag{232a}$$

$$w_e>0\qquad\forall\, e\in E \tag{233}$$

$$\Gamma^{\{\alpha\}}(\mathbf{w}) = \Gamma^{\{\alpha'\}}(\mathbf{w}) \tag{3.234}$$

$$(\mathbf{\Gamma}^{\alpha}_{\mathscr{J}} - \mathbf{\Gamma}^{\alpha'}_{\mathscr{J}})(\mathbf{w}) = 0 \tag{A.235}$$

$$\Gamma = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{236a}$$

$$E_0(\mathscr{I}, \alpha) = \mathscr{C}(U^{*\,\alpha}_{\mathscr{J}}) \cap E_0 \tag{237}$$

$$\Gamma \mathbf{w}' = \lambda_{\mathbf{w}'} \mathbf{S} \qquad \forall\, \mathbf{w}' \in \partial \mathcal{W}_{\mathfrak{m}} \tag{7.238}$$

$$\mathscr{U}_A = \{\{A\}, \{A, \sigma\}\}, \qquad \mathscr{U}_B = \{\{B\}\}, \qquad \mathscr{U}_{AB} = \{\{A, B\}, \{A, B, \sigma\}\} \tag{A.239}$$

$$\mathbb{W} = (0, 1, -1)^{\perp} \subset \mathbb{R}^3 \tag{240a}$$

$$\mathfrak{m} = \{ \bigcup_{i \in [k]} \mathscr{U}^i_{\mathscr{J}} \text{ for all } \mathscr{I} \} \tag{241}$$

$$\text{Ker } \Pi^{\intercal} = \mathbb{P} \tag{4.242}$$

$$G[U^*_{\underline{\mathscr{J}} \cup \underline{\mathscr{K}}}] = G[U^*_{\underline{\mathscr{J}}}] \oplus G[U^*_{\underline{\mathscr{K}}}] \tag{A.243}$$

$$\mathscr{I} = \mathscr{I}(X) = \underset{e \in X}{\triangle} \mathscr{K}(e) \tag{244a}$$

$$S_{\mathscr{I}} = \sum_{e \in \mathscr{C}(U^*_{\mathscr{J}})} S_{\mathscr{K}(e)} \tag{245}$$

$$\sum_{e \,\in\, \mathscr{C}^{*\alpha}_{\mathscr{I}}} S_{\mathscr{J}(e)} = \sum_{f \,\in\, \mathscr{C}^{*\beta}_{\mathscr{J}}} S_{\mathscr{K}(f)} \tag{1.246}$$

$$S'_{\mathscr{I}'} = S_{\phi(\mathscr{I}')} \tag{A.247}$$

$$\mathbb{W}' \supseteq \mathbb{W} \tag{248a}$$

$$\mathbf{S}(\widetilde{G_2}) = \lambda\,(2, 1, 1), \qquad \lambda > 0 \tag{249}$$

$$w_{A'_1 \sigma} + w_{A'_2 \sigma} = w_{B' \sigma} + w_{O' \sigma} \tag{5.250}$$

$$\Gamma = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{A.251}$$

$$\beta^{\downarrow} = \beta \tag{252a}$$

$$\operatorname{cone}\{(1,0,0),(0,1,0),(1,0,1),(0,1,1)\} \tag{253}$$

$$\operatorname{cone}\{(1,0,1),(0,1,1),(0,0,1)\} \tag{2.254}$$

$$\operatorname{cone}\{(0,1,0),(1,1,0),(0,1,1)\} \tag{A.255}$$

$$\mathscr{R}_{\text{H}} \subseteq \mathscr{R}_{\text{Q}} \subseteq \mathscr{R}_{\text{SSA}} \subseteq \mathscr{R} \tag{256a}$$

$$\Gamma = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \tag{257}$$

$$\begin{aligned} U_{AB}^{*} &= \{A,B\} \\ U_{AC}^{*} &= \{A,C\} \\ U_{BC}^{*} &= \{B,C\} \end{aligned} \tag{6.258}$$

$$\begin{aligned} e < f \quad &\Rightarrow \quad \mathscr{I}(e) \supseteq \mathscr{I}(f) \\ e, f \ \text{ incomparable} \quad &\Rightarrow \quad \mathscr{I}(e) \cap \mathscr{I}(f) = \varnothing \end{aligned} \tag{A.259}$$

$$\begin{aligned} \mathcal{W}_{\widetilde{G_2}} &= \operatorname{cone}\{(1,1,0),(0,1,1),(1,0,1)\} \\ \mathcal{W}_{\widetilde{G_2}'} &= \operatorname{cone}\{(1,1,0),(1,0,0),(1,0,1)\} \end{aligned} \tag{260a}$$

$$\forall \rho \in \mathscr{D}(\mathcal{H}),\ \mathcal{G}(\rho) \in \mathscr{F}_G. \tag{261}$$

$$\mathrm{D}\left(\rho \| \sigma\right) := \operatorname{Tr}\left[\rho(\log \rho - \log \sigma)\right], \tag{3.262}$$

$$V\left(\rho \| \sigma\right) := \operatorname{Tr}[\rho(\log \rho - \log \sigma)^2] - \mathrm{D}\left(\rho \| \sigma\right)^2. \tag{A.263}$$

$$U_g U_{g'} = U_{gg'} \tag{264a}$$

$$U_g U_{g'} = e^{i\theta(g,g')} U_{gg'}. \tag{265}$$

$$\frac{1}{|\mathcal{M}|} \sum_m |m\rangle\langle m|_M \otimes \mathcal{E}_{A\to A}^m(|\Psi\rangle\langle\Psi|_{AF}). \tag{7.266}$$

$$\sum_{m,\widehat{m}} q_{\widehat{M}M}(\widehat{m}|m) |m\rangle\langle m|_M \otimes |\widehat{m}\rangle\langle\widehat{m}|_{\widehat{M}}, \tag{A.267}$$

$$q_{\widehat{M}M}(\widehat{m}|m) := \operatorname{Tr}\left[\Gamma_{BF}^{\widehat{m}} \mathcal{E}_{A\to A}^m(|\Psi\rangle\langle\Psi|_{AF})\right]. \tag{268a}$$

$$r\left(\mathcal{C}\right) := \log |\mathcal{C}|. \tag{269}$$

$$q_{\widehat{M}M}(\widehat{m}|m) = \operatorname{Tr}\left[\Gamma_{BF}^{\widehat{m}} \mathcal{E}_{A\to A}^m(|\Psi\rangle\langle\Psi|_{AF})\right] \geq 0 \tag{4.270}$$

$$C_{\mathfrak{E},\mathfrak{D}}(\Psi_{AF}) \leq C^{\dagger}_{\mathfrak{E},\mathfrak{D}}(\Psi_{AF}) \tag{A.271}$$

$$\Psi_{AF} = \sum_{k\in\mathcal{K}} \sqrt{P_K(k)}\Psi_{AF,k}, \tag{272a}$$

$$\xi_{AF} := (\mathcal{G}_A \otimes \mathrm{id}_F)(\Psi_{AF}) = \sum_{k\in\mathcal{K}} P_K(k)\pi_k \otimes \rho_{F,k}, \tag{273}$$

$$\mathrm{H}(\xi_{AF}) = \mathrm{H}(K)_\xi + \mathrm{H}(A|K)_\xi + \mathrm{H}(F|K)_\xi = \mathrm{H}(A)_\xi + \mathrm{H}(F|K)_\xi, \tag{1.274}$$

$$\mathrm{D}\left(\Psi_{AF}\|\Delta_A(\Psi_{AF})\right) = \mathrm{H}(\Delta(\Psi_A)), \tag{A.275}$$

$$|\Psi\rangle_{AF} = \sum_{n=0}^{\infty} \sqrt{N^n/(N+1)^{n+1}}\, |n\rangle_A |n\rangle_F . \tag{276a}$$

$$T := \sum_m \sum_x |m\rangle\langle m|_M \otimes |x\rangle\langle x|_X \otimes \Gamma^{m|x} \tag{277}$$

$$|\Psi\rangle_{AF} := \frac{1}{\sqrt{N}} \sum_{n\in\mathcal{N}} |\psi_{A,n}\rangle |n\rangle_F , \tag{5.278}$$

$$\overline{\rho}_A := \mathcal{G}(\Psi_A) = \frac{1}{N}\sum_{n\in\mathcal{N}} \mathcal{G}(\psi_{A,n}) = \frac{1}{N}\sum_{n\in\mathcal{N}} \overline{\rho}_{A,n}. \tag{A.279}$$

$$\xi_{AF} = \frac{1}{N}\sum_{n\in\mathcal{N}} \overline{\rho}_{A,n} \otimes |n\rangle\langle n|_F. \tag{280a}$$

$$\Pi^{f,t} := \sum_{n\in f^{-1}(t)} |n\rangle\langle n|_F. \tag{281}$$

$$\delta^{f,t} := 1 - \frac{1}{L}\sum_{n\in f^{-1}(t)} \mathrm{Tr}\left[Q^{f,t}_{A,n}\overline{\rho}_{A,n}\right]. \tag{2.282}$$

$$|\Psi^{f,t}_{AF}\rangle := \frac{1}{\sqrt{L}} \sum_{n\in f^{-1}(t)} |\psi_{A,n}\rangle |n\rangle_F . \tag{A.283}$$

$$\langle\Psi^{f,t}_{AF}|\boldsymbol{Q}^{f,t}_{AF}|\Psi^{f,t}_{AF}\rangle = \frac{1}{L}\sum_{n\in f^{-1}(t)} \langle\psi_{A,n}|Q^{f,t}_{A,n}|\psi_{A,n}\rangle = 1 - \delta^{f,t}. \tag{284a}$$

$$|\boldsymbol{b}^{l'}_F\rangle := \frac{1}{\sqrt{L}} \sum_{l=1}^{L} \zeta^{ll'} |l\rangle_F , \; l' = 1, \cdots, L. \tag{285}$$

$$\begin{aligned} \rho_{MA} &:= \frac{1}{|\mathcal{M}|}\sum_m |m\rangle\langle m|_M \otimes \mathcal{E}^m(\Psi_A), \\ \sigma_{MA} &:= \pi_M \otimes \mathcal{G}(\Psi_A), \end{aligned} \tag{6.286}$$

$$T := \sum_m |m\rangle\langle m|_M \otimes \Gamma^m \tag{A.287}$$

$$a = \begin{pmatrix} a^1 \\ a^2 \\ \vdots \\ a^k \end{pmatrix}, \quad b = (b_1, b_2, \dots, b_k). \tag{288a}$$

$$(F \circ L[V^{-1}])^* = F^* \circ R[V] \tag{289}$$

$$J^{i,j,*}(\theta) := \frac{\partial^2 F^*}{\partial \eta_i \partial \eta_j}(\eta) \tag{3.290}$$

$$\gamma(\hat{\Theta}|\Theta) J^*(\theta_1) \geq J^*(\theta_2) \tag{A.291}$$

$$u \cdot \nabla F(\theta) > D, \tag{292a}$$

$$\lim_{n\to\infty} D^F(\theta^{(n)} \| \theta) = \inf_{\theta' \in \mathcal{M}} D^F(\theta' \| \theta). \tag{293}$$

$$\partial \mathcal{M}_\lambda = \cup_{\lambda' \in \Lambda_\lambda} \mathcal{M}_{\lambda'} \tag{7.294}$$

$$D(q\|p) = \sum_\omega p(\omega)(\log p(\omega) - \log q(\omega)). \tag{A.295}$$

$$\tau(t\xi) \to \left(\sum_{i=1}^{l'} \frac{\exp\left(\sum_{j=1}^{d-l} \theta^j f_j(x_i) \right)}{\sum_{i'=1}^{l'} \exp\left(\sum_{j=1}^{d-l} \theta^j f_j(x_{i'}) \right)} f_{d-j}(x_i) \right)_{j=1}^{l} \tag{296a}$$

$$\tau_{(\theta^1,\dots,\theta^{d-l})}(\xi) := \left(\frac{\operatorname{Tr} X_{d-j} \exp\left(\sum_{i=1}^{d} \theta^j X_{d-i} \right)}{\mu(\theta)} \right)_{j=1}^{l} \tag{297}$$

$$u_{j'}^{i'} \Big(\eta_{i'}(\bar{\theta}^{(t+1)}) + \sum_{i=1}^{d} J_{i',i}(\bar{\theta}^{(t+1)}) \sum_{j=k+1}^{d} u_j^i \kappa^j \Big) = a_{j'} \tag{4.298}$$

$$D^F(\theta^{(t_2)} \| \bar{\theta}^{(t_2)}) \lessapprox \epsilon_1 \tag{A.299}$$

$$\sum_{x\in\mathcal{X}, y\in\mathcal{Y}} d(x,y) W \times P_X(x,y) \leq D \tag{300a}$$

$$\begin{aligned} I(X;Y)_{W_* \times P_X} =& D(W_* \times P_X \| Q_{Y,1} \times P_X) \\ \geq& D(W_1 \times P_X \| Q_{Y,1} \times P_X) \geq I(X;Y)_{W_1 \times P_X}, \end{aligned} \tag{301}$$

$$\begin{aligned} \bar{\theta}^j =& \theta^j, \quad j = 1, \dots, n_1(n_2-1) - 1 \\ \bar{\theta}^{n_1(n_2-1)} =& \sum_{x\in\mathcal{X}} \sum_{y=1}^{n_2-1} d(x,y) \theta^{(x-1)(n_2-1)+y}. \end{aligned} \tag{1.302}$$

$$0 \leq \tau_k[P_Y] - \tau_*[P_Y] \leq -\frac{1}{2^k} \frac{\frac{d}{d\tau}\hat{F}[P_Y](0)}{\zeta_-}. \tag{A.303}$$

$$P_{XY|k}[P_Y] := (1-\kappa[P_Y])P_{X,Y|\tau_k[P_Y]}[P_Y] + \kappa[P_Y]P_X \times P_Y. \tag{304a}$$

$$\begin{aligned} \hat{F}[P_Y](\tau_k[P_Y]) - \hat{F}[P_Y](\tau_*[P_Y]) \leq & \epsilon' \\ D(P_{XY|k}[P_Y] \| P_{X,Y|\tau_k[P_Y]}[P_Y]) \leq & \epsilon'. \end{aligned} \tag{305}$$

$$\frac{1}{t_1 - 1} D(\theta_* \| \theta_1) \leq \frac{\epsilon}{3}. \tag{5.306}$$

$$I(X;Y)_{P^*_{XY}} := 0.100039, \tag{A.307}$$

$$P^*_Y = \begin{pmatrix} 0.185555 \\ 0.288401 \\ 0.526045 \end{pmatrix}. \tag{308a}$$

$$P_{XY,\theta,\tau}(x,y) := \frac{P_{\theta,Y}(y) P_X(x) e^{\sum_{i=0}^{n_1^{-1}} g_i(x,y)\tau_i}}{\sum_{x'y'} P_{\theta,Y}(y') P_X(x') e^{\sum_{i=0}^{n_1^{-1}} g_i(x',y')\tau_i}}, \tag{309}$$

$$\operatorname{Tr} \Delta \rho_R \otimes \sigma_B \leq D \tag{2.310}$$

$$I(A;B|E)_{\rho_{ABE}} := H(AE)_{\rho_{ABE}} + H(BE)_{\rho_{ABE}} - H(ABE)_{\rho_{ABE}} - H(E)_{\rho_{ABE}}. \tag{A.311}$$

$$M_k(\eta) := \eta_{d_A^2 k} I + \sum_{i=1}^{d_A^2 - 1} X_i \eta_{d_A^2(k-1)+i} \tag{312a}$$

$$\|\nabla F(x) - \nabla F(y)\| \leq L \|x - y\| \tag{313}$$

$$\infty = \sum_{t=1}^{\infty} c(\frac{1}{t}) \leq \sup_{\theta \in \mathcal{E}} D^F(\theta \| \theta_{(1)}), \tag{6.314}$$

$$\begin{aligned} \sum_{x,y} P_X(x) W_{\theta,x}(y) f_j(y) = c_{1,j} \\ \sum_{x,y} P_X(x) W_{\theta,x}(y) \delta_{i,x} = c_{2,i}, \end{aligned} \tag{A.315}$$

$$\eta_{2,0}(\theta, \tau(\theta)) = D, \tag{316a}$$

$$P_2 J_{\theta_0,\tau(\theta_0),3} P_1 v_1 + P_2 J_{\theta_0,\tau(\theta_0),3} P_2 v_2 = 0. \tag{317}$$

$$\begin{aligned} H(P) := & -\sum_{x \in \mathcal{X}} P(x) \log P(x), \\ D(P\|Q) := & \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}. \end{aligned} \tag{3.318}$$

$$D(W_x \| P_{\theta,Y}) = \phi(\theta) - H(W_{n_1}). \tag{A.319}$$

$$\begin{aligned}\theta^j =&\frac{1}{f_j(y_j)-f_j(y_j')}\Big(\sum_{i=1}^{n_1-1}(f_i(y_j')-f_i(y_j))\theta^i\\&+\log\frac{-f_j(y_j')}{f_j(y_j)}\Big)\end{aligned}\tag{320a}$$

$$W_1:=\begin{pmatrix}1-p\\p\end{pmatrix},\quad W_2:=\begin{pmatrix}1-q\\q\end{pmatrix}.\tag{321}$$

$$\begin{aligned}C(W)=&\phi(\theta)-h(p)\\=&\frac{ph(q)-qh(p)}{q-p}+\log\Big(1+e^{\frac{-(q+p)(h(q)-h(p))}{q-p}}\Big),\end{aligned}\tag{7.322}$$

$$P_{\theta,Y}=\begin{pmatrix}e^{\frac{q(h(q)-h(p))}{q-p}-\phi(\theta)}\\e^{\frac{-p(h(q)-h(p))}{q-p}-\phi(\theta)}\end{pmatrix}.\tag{A.323}$$

$$\begin{aligned}H(W_1)=&H(W_2)=h(\epsilon)\\H(W_3)=&\log 2,\quad H(W_4)=h(2\epsilon)+(1-2\epsilon)\log 2.\end{aligned}\tag{324a}$$

$$\theta_4=\begin{pmatrix}h_{4,\epsilon}\\h_{4,\epsilon}\end{pmatrix}\tag{325}$$

$$P_{4,Y}=\begin{pmatrix}e^{-\phi_4(\theta_4)}\\e^{-\phi_4(\theta_4)}\\e^{\frac{h_{4,\epsilon}}{\epsilon}-\phi_4(\theta_4)}\end{pmatrix}.\tag{4.326}$$

$$\theta_3=\begin{pmatrix}h_{3,\epsilon}\\h_{3,\epsilon}\end{pmatrix}\tag{A.327}$$

$$P_{3,Y}=\frac{1}{2+e^{-\frac{h_{3,\epsilon}}{\epsilon}}}\begin{pmatrix}1\\1\\e^{-\frac{h_{3,\epsilon}}{\epsilon}}\end{pmatrix}.\tag{328a}$$

$$\theta_1=\begin{pmatrix}h_{3,\epsilon}\\h_{1,\epsilon}\end{pmatrix}\tag{329}$$

$$\widehat{Q}_{1,X}=V_1^{-1}P_{1,Y}=e^{h_{1,\epsilon}-\phi_1(\theta_1)}\begin{pmatrix}\kappa_1\\\kappa_2\\\kappa_3\end{pmatrix},\tag{1.330}$$

$$\begin{aligned}&h(2\epsilon p)-ph(2\epsilon)\\=&(1-2\epsilon p)\log 2+h(2\epsilon p)\\&-(1-p)\log 2-p(h(2\epsilon)+(1-2\epsilon)\log 2).\end{aligned}\tag{A.331}$$

$$D(W_x\|\rho_\theta)=\phi(\theta)-H(W_{n_2^2})\tag{332a}$$

$$\operatorname{Tr}W_i\sum_{j=1}^{n_2^2-1}A_j\theta^j=\sum_{j=1}^{n_2^2-1}h_{i,j}\theta^j=-H(W_i)+H(W_{n_2^2}).\tag{333}$$

$$D(W_x\|\rho_\theta) = -H(W_x) - \sum_{j=1}^{n_2^2-1} \theta^j h_{x,j} - \phi(\theta). \tag{5.334}$$

$$\begin{aligned} &- H(W_x) - \sum_{j=1}^{n_2^2-1} \theta^j h_{x,j} - \phi(\theta) \\ =& - H(W_{n_2^2}) - \sum_{j=1}^{n_2^2-1} \theta^j h_{n_2^2} - \phi(\theta) = -H(W_{n_2^2}) - \phi(\theta) \end{aligned} \tag{A.335}$$

$$C = \log \Big(\sum_{x'=1}^{n_1} G_{x'} \Big), \tag{336a}$$

$$G_{x'} := \sum_{x,y} g_{s,x'} W(y|x) \log W(y|x). \tag{337}$$

$$P_{\theta,X}(x) := e^{\sum_{j=1}^{l} \theta^j f_j(x) - \phi(\theta)}, \tag{2.338}$$

$$D(P_{X,1}\|P_{X,2}) = D(P_{X,1}\|P_{X,*}) + D(P_{X,*}\|P_{X,2}). \tag{A.339}$$

$$D(W_{x_0}\|W \cdot Q_X) < D(W_{x_0}\|W \cdot Q_{X,\epsilon}). \tag{340a}$$

$$\sum_{y\in\mathcal{Y}} W_x(y) f_j(y) = 0, \quad \sum_{y\in\mathcal{Y}} W_{n_1}(y) f_{j'}(y) = 0 \tag{341}$$

$$h_{x,j} := \sum_{y\in\mathcal{Y}} W_x(y) f_j(y) \tag{6.342}$$

$$D(W_x\|P_{\theta,Y}) = -H(W_x) - \sum_{j=1}^{n_1-1} \theta^j h_{x,j} - \phi(\theta_1, \theta_2(\theta_1)). \tag{A.343}$$

$$Q_{Y,t}(y) := Q_{Y,0}(y) e^{t f_{x_0}(y) - \varphi(y)}, \tag{344a}$$

$$\varphi(y) := \log \sum_{y\in\mathcal{Y}} Q_{Y,0}(y) e^{t f_{x_0}(y)}. \tag{345}$$

$$C(W) = \sum_{x\in\mathcal{X}\setminus\{x_0\}} Q_{X,1}(x) D(W_x\|W \cdot Q_{X,1}). \tag{3.346}$$

$$\sum_j f_j a^j_{j'} = \overline{f}_{j'}. \tag{A.347}$$

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & \gamma \\ \gamma^* & 1 \end{pmatrix}, \tag{348a}$$

$$P(|\tilde{\Psi}\rangle, |\Psi\rangle) = \mathrm{Tr}\{\rho\, |\Psi\rangle\langle\Psi|\} = \frac{1}{2}(1 + \mathrm{Re}(\gamma)) = \frac{1}{2}(1 + e^{-s}\cos\phi)\,. \tag{349}$$

$$\omega_{\mathbf{q}} = qv\cos\theta_{\mathbf{qv}} - \frac{q^2}{2m_\chi}\ , \tag{7.350}$$

$$F_A(q) = \frac{3j_1(qr_A)}{qr_A}e^{-q^2s_p^2/2}\ , \tag{A.351}$$

$$F(x) = \frac{3j_1(x)}{x}\ . \tag{352a}$$

$$m_\chi^{\rm knee} \sim \frac{1}{R_{\phi\chi}\sqrt{\Delta x\, r_C}}. \tag{353}$$

$$\sum_f |\langle f|\mathcal{F}_T(\mathbf{q})|i\rangle|^2 g(\mathbf{q},\omega_1) = a_0^6\frac{q^2}{2}\int\frac{d^3\mathbf{k}_1}{(2\pi)^3}\omega_1 S(\mathbf{q}-\mathbf{k}_1)g(\mathbf{q},\omega_1)\ , \tag{4.354}$$

$$\mathcal{L} \supset \frac{a^2}{8f_a^2}\sum_{N=p,n}\delta m_N\,\bar{N}N\ , \tag{A.355}$$

$$\bar{\sigma} = \frac{\mu^2}{256\pi}\left(\frac{\delta m_N}{f_a^2}\right)^2\frac{1}{m_a^2}\ , \tag{356a}$$

$$V(r) = -\frac{y_n^2}{4\pi}\frac{1}{r}e^{-m_\phi r}, \tag{357}$$

$$\Delta N_{\rm eff} = \frac{4}{7}\sum_i g_i\left(\frac{g(T_{\nu_L}^{\rm dec})}{g(T_i)}\right)^{4/3}\ , \tag{1.358}$$

$$V_{\rm BEC} \simeq \frac{1}{2}m\omega^2 r^2, \tag{A.359}$$

$$\Psi_0(r) = \left(\frac{m\omega}{\pi\hbar}\right)^{3/4}\exp\left(-\frac{m}{2\hbar}\omega r^2\right)\ . \tag{360a}$$

$$F(\mathbf{q}) = \exp\left[-\left(\frac{q\,r_{\rm cloud}}{2}\right)^2\right]\ , \tag{361}$$

$$r_{\rm cloud} = \left(\frac{\hbar}{m\omega}\right)^{1/2}\ . \tag{5.362}$$

$$n_{-\mathbf{k}_2} = \frac{1}{\sqrt{V}}\sum_{i=1,\cdots,N_A}e^{-i\mathbf{k}_2\cdot\mathbf{y}_i}, \tag{A.363}$$

$$\frac{d\mathbf{h}}{dt} = \sigma(W\mathbf{h})\odot[-\mathbf{h}+J\phi(\mathbf{h})] \tag{364a}$$

$$\dot{h}_i = \sum_j M_{ij}(\mathbf{h})G_j(\mathbf{h}) \tag{365}$$

$$\sigma_{notch}(x) = \begin{cases} 0 & 0 \leq |x| \leq \beta/2 \\ 1 & \text{o/w} \end{cases}, \tag{2.366}$$

$$\int Dx \left(\phi'\left(\sqrt{\Delta_h}x\right)\right)^2 < 1/\mu_\sigma, \tag{A.367}$$

$$\sigma_\beta^{DMFT} = \frac{C_\phi(\Delta)}{2}, \tag{368a}$$

$$\mathcal{D}_{ij} = \partial \dot{h}_i/\partial h_j = \sigma_i \left(-\delta_{ij} + J_{ij}(\phi'(h_j)r_j(t)\right). \tag{369}$$

$$\Phi(\Delta) = \int Dx \phi(\sqrt{\Delta}x)^2, \tag{6.370}$$

$$\gamma^2 \mu_\sigma (1-\mu_\sigma)(1+(G-1)\mu_\sigma)g^2 \geq 1. \tag{A.371}$$

$$R_{ssRNN} = G^2 \frac{\mu_\sigma^2}{(1-\mu_\sigma)^2} R_{gRNN} = G^2 R_{gRNN}^{-1}. \tag{372a}$$

$$g_{\mu\nu} = \delta_{\mu\nu} + \Gamma_{\mu\nu}\phi'(f_\mu)\phi'(f_\nu) \tag{373}$$

$$\tilde{n}_i^{(j)} = \frac{\partial G_j}{\partial h^i} \equiv G_{j,i}. \tag{3.374}$$

$$\frac{\partial \Phi}{\partial h^i} = \Phi_{,i}. \tag{A.375}$$

$$n_i = \frac{\Phi_{,i}}{\sqrt{g^{ij}\Phi_{,i}\Phi_{,j}}}. \tag{376a}$$

$$\mathcal{D}_{ij} = \frac{\partial G_i}{\partial h_j} = -\delta_{ij} + J_{ij}\phi'(h_j). \tag{377}$$

$$\Phi_{,j} = \sum_{i=1}^{N_a} c_i G_i \mathcal{D}_{ij}. \tag{7.378}$$

$$\frac{\partial n_i}{\partial h_j} = \kappa_i \delta_{ij} - n_i n_j \kappa_j, \tag{A.379}$$

$$\kappa_i = \frac{1}{\left(\hat{l}^T \mathcal{D}\mathcal{D}^T \hat{l}\right)^{1/2}} \sum_k \hat{l}_k J_{ki} \phi''(h_i), \tag{380a}$$

$$\frac{\partial \mathcal{D}_{ki}}{\partial h_j} = J_{ki}\phi''(h_i)\delta_{ij}. \tag{381}$$

$$e_\mu^i = \frac{\partial h^i}{\partial f^\mu}. \tag{4.382}$$

$$\sum_{i=1}^{N} n_i e_\mu^i = 0. \tag{A.383}$$

$$\langle K_{\mu\nu} \rangle = 0, \tag{384a}$$

$$\mathcal{N} \equiv \langle \hat{l}^T \mathcal{D} \mathcal{D}^T \hat{l} \rangle, \approx 1 + \frac{1}{N} \sum_{i=1}^{N} \phi'(h_i)^2. \tag{385}$$

$$K_{\mu\nu} \approx \left[\kappa_\mu + \kappa \, \Gamma \left(\phi'(f_\mu) \right)^2 \right] \delta_{\mu\nu} \tag{1.386}$$

$$\begin{aligned} \frac{d\mathbf{h}}{dt} &= \sigma_{notch}(W\mathbf{h}) \odot [-\mathbf{h} + J\phi(\mathbf{h})] \\ \sigma_{notch}(x) &= \begin{cases} 0 & 0 \leq |x| \leq \beta/2 \\ 1 & \text{o/w} \end{cases} \end{aligned} \tag{A.387}$$

$$\begin{aligned} \tau \dot{r} &= -r + J s_E - J s_I + I, \\ \tau_E \dot{s}_E &= -s_E + \phi(r), \\ \tau_I \dot{s}_I &= -s_I + \phi(r). \end{aligned} \tag{388a}$$

$$\begin{aligned} L[s_E - s_I](u) &\approx (\tau_I - \tau_E) u L[\phi(r)] \\ &= -\Delta\tau L \left[\frac{d\phi(r)}{dt} \right], \quad \Delta\tau = \tau_E - \tau_I. \end{aligned} \tag{389}$$

$$\tau \dot{r} \approx -r - \Big(J \Delta\tau \phi'(r) \Big) \dot{r} + I, \tag{5.390}$$

$$A_{ij} = \begin{cases} 1 & i \in \partial j \\ 0 & \text{otherwise}\,, \end{cases} \tag{A.391}$$

$$\Pi_{ij} = \frac{A_{ij}}{k_i}\,, \tag{392a}$$

$$\rho_i = \frac{k_i}{\sum_{j \in V} k_j}\,, \tag{393}$$

$$\rho_i \Pi_{ij} = \rho_j \Pi_{ji} \tag{2.394}$$

$$C_n = \frac{1}{n} \sum_{\ell=1}^{n} f(X_\ell)\,, \tag{A.395}$$

$$\mathbb{P}(C_n = c) = e^{-nI(c) + o(n)}\,, \tag{396a}$$

$$I(c) = -\lim_{n \to \infty} \frac{1}{n} \log \mathbb{P}(C_n = c)\,. \tag{397}$$

$$C_n = \sum_{i,j \in V} f(i) L_n^{(2)}(i,j)\,. \tag{6.398}$$

$$\lambda_{s,N}[\nu^*] = \lim_{n\to\infty} \frac{1}{n}\mathbb{E}\left[e^{nsC_n}\right] \ , \tag{A.399}$$

$$I(c) = \sup_{s\in\mathbb{R}}\{sc - \lambda_{s,N}[\nu^*]\}\,. \tag{400a}$$

$$I(c) = -\lambda_{1,N}[\nu^*] - \lambda_{2,N}[\nu^*] - \lambda_{4,N}[\nu^*]\,, \tag{401}$$

$$c = \frac{d\lambda_{s,N}[\nu]}{ds}\,. \tag{3.402}$$

$$(\Pi_s)_{ij} = \frac{\nu^*_{ij}(s)}{\sum_{k\in V}\nu^*_{ik}(s)}\,. \tag{A.403}$$

$$C_n = \frac{1}{n}\sum_{\ell=1}^{n}\frac{k_{X_\ell}}{\bar{k}}\,, \tag{404a}$$

$$\lambda_{s,N}[\nu] = h(s,N,\nu_{12},\nu_{23},\nu_{34},\nu_{44}) + \epsilon(1-2\nu_{12}-2\nu_{23}-2(N-3)\nu_{34}-(N-3)(N-4)\nu_{44})\,, \tag{405}$$

$$\lambda_{s,\infty} = \begin{cases} -\frac{\log 2}{2} & s \le -\frac{\log 2}{2} \\ s & s > -\frac{\log 2}{2} \end{cases} \tag{7.406}$$

$$\frac{d\lambda_{s,\infty}}{ds} = \begin{cases} 0 & s \le -\frac{\log 2}{2} \\ 1 & s > -\frac{\log 2}{2}\,, \end{cases} \tag{A.407}$$

$$I(c) = \begin{cases} \frac{\log 2}{2} - c\frac{\log 2}{2} & 0 \le c \le 1 \\ \infty & \text{otherwise}\,, \end{cases} \tag{408a}$$

$$s = -\frac{\log 2}{2} + \frac{\tilde{s}}{N}\,, \tag{409}$$

$$\epsilon = \frac{\log 2}{2} - \frac{\tilde{\epsilon}_{\tilde{s}}}{N}\,, \tag{4.410}$$

$$\lambda_{\tilde{s},N} \approx -\frac{\log 2}{2} + \frac{\tilde{\epsilon}_{\tilde{s}}}{N}\,, \tag{A.411}$$

$$\tilde{\epsilon}_{\tilde{s}} = \frac{1}{8}\left(\sqrt{2} + 4\tilde{s} - 7\log 2 + \sqrt{2 + 16\sqrt{2} + 2\sqrt{2}\log 2 - \log^2 2 - 8\sqrt{2}\tilde{s} - 8\log 2\tilde{s} + 16\tilde{s}^2}\right)\,, \tag{412a}$$

$$C_n = \frac{1}{n}\sum_{\ell=1}^{n}\frac{X_\ell}{b}\,, \tag{413}$$

$$\lambda_{s,b} = -\epsilon = \log\left[\frac{1}{4b}\left(be^{\frac{s}{b}} + 2(b-1)e^s + \sqrt{4(b-1)^2e^{2s} + b^2e^{\frac{2s}{b}} - 4(b-3)be^{\frac{bs+s}{b}}}\right)\right]\,. \tag{1.414}$$

$$\lambda_{s,\infty} = \begin{cases} -\log 2 & s \le -\log 2 \\ s & s > -\log 2 \end{cases} \tag{A.415}$$

$$\frac{d\lambda_{s,\infty}}{ds} = \begin{cases} 0 & s \leq -\log 2 \\ 1 & s > -\log 2\,. \end{cases} \tag{416a}$$

$$I(c) = \begin{cases} \log 2 - c\log 2 & 0 \leq c \leq 1 \\ \infty & \text{otherwise}\,. \end{cases} \tag{417}$$

$$\Pi_{s^-} = \begin{pmatrix} 1 + O(b^{-1}) & O(b^{-1}) \\ 1 - 2e^s + O(b^{-1}) & 2e^s + O(b^{-1}) \end{pmatrix} \tag{5.418}$$

$$\Pi_{s^+} = \begin{pmatrix} \frac{1}{2e^s} + O(b^{-1}) & 1 - \frac{1}{2e^s} + O(b^{-1}) \\ O(b^{-1}) & 1 + O(b^{-1}) \end{pmatrix}. \tag{A.419}$$

$$\lambda_{\tilde{s},b} \approx -\log 2 + \frac{\tilde{\epsilon}_{\tilde{s}}}{2\sqrt{b}}\,, \tag{420a}$$

$$u^{e_Y}_{\mathbf{k},\mathbf{Q},\alpha,\eta} = \frac{1}{\sqrt{2}}(u^{n=+}_{\mathbf{k},\mathbf{Q},\alpha,\eta} + ie_Y u^{n=-}_{\mathbf{k},\mathbf{Q},\alpha,\eta}) \tag{421}$$

$$V^{\eta,e_Y,e'_Y}_{\mathbf{k},\mathbf{k}'} = \frac{1}{N_0\omega_{A_1}} \sum_{\mathbf{G}_M,l} G^{\eta,e_Y,l}_{\mathbf{k},\mathbf{k}',-l\eta\mathbf{q}_2+\mathbf{G}_M} G^{-\eta,e'_Y,l}_{-\mathbf{k},-\mathbf{k}',l\eta\mathbf{q}_2-\mathbf{G}_M} \tag{2.422}$$

$$D_{AB}(\delta_3^{nn}) = -\begin{pmatrix} K_{0t} & 0 \\ 0 & K_{0r} \end{pmatrix}. \tag{A.423}$$

$$D_{AA}(\delta_1^{nnn}) = -\begin{pmatrix} K_{r1} & K_2 \\ -K_2 & K_{t1} \end{pmatrix}. \tag{424a}$$

$$\mathbf{b}_{M1} = \mathbf{q}_2 - \mathbf{q}_1 = k_\theta(\frac{\sqrt{3}}{2}, \frac{3}{2}), \qquad \mathbf{b}_{M2} = \mathbf{q}_3 - \mathbf{q}_1 = k_\theta(-\frac{\sqrt{3}}{2}, \frac{3}{2}). \tag{425}$$

$$T_j = w_0\sigma_0 + w_1\sigma_x \cos\frac{2\pi(j-1)}{3} + w_1\sigma_y \sin\frac{2\pi(j-1)}{3}\ . \tag{6.426}$$

$$T_{\mathbf{R}} c^\dagger_{\mathbf{k},\mathbf{Q},\alpha,\eta,s} T^{-1}_{\mathbf{R}} = e^{-i\mathbf{k}\cdot\mathbf{R}} c^\dagger_{\mathbf{k},\mathbf{Q},\alpha,\eta,s}\ , \tag{A.427}$$

$$\hat{g} c^\dagger_{\mathbf{k},\mathbf{Q},\eta,\alpha,s} \hat{g}^{-1} = \sum_{\mathbf{Q}'\alpha'\eta'} c^\dagger_{\hat{g}\mathbf{k},\mathbf{Q}',\eta',\alpha',s} [D(\hat{g})]_{\mathbf{Q}'\eta'\alpha',\mathbf{Q}\eta\alpha}\ . \tag{428a}$$

$$[D(\hat{C})]_{\mathbf{Q}'\alpha'\eta',\mathbf{Q}\alpha\eta} = \delta_{\mathbf{Q}',\mathbf{Q}} [\sigma_z]_{\alpha'\alpha} [\tau_0]_{\eta'\eta}\ . \tag{429}$$

$$C^2 = 1, \qquad [C,T] = 0, \qquad [C,C_{3z}] = 0, \qquad \{C,C_{2x}\} = 0, \qquad \{C,C_{2z}T\} = 0, \qquad [C,P] = 0\ . \tag{3.430}$$

$$\sum_{\mathbf{Q}',\beta} [h^{(\eta)}_{\mathbf{Q},\mathbf{Q}'}(\mathbf{k})]_{\alpha\beta} u_{\mathbf{Q}'\beta;n\eta}(\mathbf{k}) = \epsilon_{n,\eta}(\mathbf{k}) u_{\mathbf{Q}\alpha;n\eta}(\mathbf{k})\ , \tag{A.431}$$

$$[D(\hat{C}_{2z})]_{\eta\eta'} u_{n\eta'}(\mathbf{k}) = \sum_m [B^{C_{2z}}(\mathbf{k})]_{m\eta,n\eta'} u_{m\eta}(-\mathbf{k})\ , \quad [D(\hat{T})]_{\eta\eta'} u^*_{n\eta'}(\mathbf{k}) = \sum_m [B^T(\mathbf{k})]_{m\eta,n\eta'} u_{m\eta}(-\mathbf{k})\ , \tag{432a}$$

$$[D(\hat{P})]_{\eta\eta'}u_{n\eta'}(\mathbf{k}) = \sum_m [B^P(\mathbf{k})]_{m\eta,n\eta'}u_{m\eta}(-\mathbf{k}) \ . \tag{433}$$

$$\hat{g}\gamma^\dagger_{\mathbf{k},n,\eta',s}\hat{g}^{-1} = \sum_{m\eta}[B^g(\mathbf{k})]_{m\eta,n\eta'}\gamma^\dagger_{\hat{g}\mathbf{k},m,\eta,s} \ . \tag{7.434}$$

$$[D(\hat{C}_{2z})D(\hat{T})]_{\eta\eta'}u^*_{n\eta'}(\mathbf{k}) = \sum_m [B^{C_{2z}T}(\mathbf{k})]_{m\eta,n\eta'}u_{m\eta}(\mathbf{k}) \ , \tag{A.435}$$

$$[D(\hat{P})D(\hat{C}_{2z})]_{\eta\eta'}u_{n\eta'}(\mathbf{k}) = \sum_m [B^{C_{2z}P}(\mathbf{k})]_{m\eta,n\eta'}u_{m\eta}(\mathbf{k}) \ . \tag{436a}$$

$$[B^{C_{2z}T}(\mathbf{k})]_{m\eta,n\eta'} = \delta_{\eta,\eta'}\delta_{m,n}e^{i\varphi^{C_{2z}T}_{n,\eta'}(\mathbf{k})} \ , \qquad [B^{C_{2z}P}(\mathbf{k})]_{m\eta,n\eta'} = \delta_{-\eta,\eta'}\delta_{-m,n}e^{i\varphi^{C_{2z}P}_{n,\eta'}(\mathbf{k})} \ , \tag{437}$$

$$B^{C_{2z}T}(\mathbf{k})B^{C_{2z}T*}(\mathbf{k}) = [B^{C_{2z}P}(\mathbf{k})]^2 = I \ , \qquad B^{C_{2z}P}(\mathbf{k})B^{C_{2z}T}(\mathbf{k}) = B^{C_{2z}T}(\mathbf{k})B^{C_{2z}P*}(\mathbf{k}) \ . \tag{4.438}$$

$$[B^{C_{2z}T}(\mathbf{k})]_{m\eta,n\eta'} = \delta_{\eta,\eta'}\delta_{m,n} \ , \qquad [B^{C_{2z}P}(\mathbf{k})]_{m\eta,n\eta'} = -\mathrm{sgn}(n)\eta'\delta_{-\eta,\eta'}\delta_{-m,n} \ . \tag{A.439}$$

$$(C_{2z}T)\gamma^\dagger_{\mathbf{k},n,\eta,s}(C_{2z}T)^{-1} = \gamma^\dagger_{\mathbf{k},n,\eta,s} \ , \qquad (C_{2z}P)\gamma^\dagger_{\mathbf{k},n,\eta,s}(C_{2z}P)^{-1} = -\mathrm{sgn}(n)\eta\gamma^\dagger_{\mathbf{k},-n,-\eta,s} \ . \tag{440a}$$

$$\lim_{\mathbf{q}\to\mathbf{0}} f_{n,\eta}(\mathbf{k}+\mathbf{q},\mathbf{k}) = 0 \tag{441}$$

$$B^{C_{2z}T}(\mathbf{k}) = \zeta^0\tau^0 \ , \qquad B^{C_{2z}P}(\mathbf{k}) = -\zeta^y\tau^y \ . \tag{1.442}$$

$$B^{C_{2z}}(\mathbf{k}) = \zeta^0\tau^x \ , \qquad B^T(\mathbf{k}) = \zeta^0\tau^x \ , \qquad B^P(\mathbf{k}) = -i\zeta^y\tau^z \ . \tag{A.443}$$

$$\gamma^{(n_\mathrm{B})\dagger}_{\mathbf{k},e_Y,\eta,s} = \frac{\gamma^\dagger_{\mathbf{k},n_\mathrm{B},\eta,s} + ie_Y\gamma^\dagger_{\mathbf{k},-n_\mathrm{B},\eta,s}}{\sqrt{2}} \ , \quad (e_Y = \pm 1). \tag{444a}$$

$$u'_{e_Y,n_\mathrm{B},\eta}(\mathbf{k}) = \frac{u_{+n_\mathrm{B},\eta}(\mathbf{k}) + ie_Y u_{-n_\mathrm{B},\eta}(\mathbf{k})}{\sqrt{2}} \ . \tag{445}$$

$$\mathbf{A}(\mathbf{k}) = \begin{pmatrix} 0 & i\mathbf{a}(\mathbf{k}) \\ -i\mathbf{a}(\mathbf{k}) & 0 \end{pmatrix}. \tag{5.446}$$

$$e_{2,n_\mathrm{B}} = \frac{1}{2\pi}\sum_i \oint_{\partial D_i} d\mathbf{k}\cdot\mathbf{a}(\mathbf{k}) = \frac{1}{2\pi}\int_{\mathrm{MBZ}-\sum_i D_i} d^2\mathbf{k}\ \Omega(\mathbf{k}) \ , \tag{A.447}$$

$$B^{P\prime}_{e_Y,e'_Y}(\mathbf{k}) = u'^\dagger_{e_Y,n_B,+}(-\mathbf{k})D(P)u'_{e_Y,n_B,+}(\mathbf{k}) = -i\chi(\mathbf{k})e_Y\delta_{e_Y,e'_Y}. \tag{448a}$$

$$(-1)^{C^{n_\mathrm{B}}_{e_Y,\eta,s}} = \prod_K \chi(K), \tag{449}$$

$$B^C(\mathbf{k}) = \zeta^y\tau^z \ . \tag{2.450}$$

$$T = 3p - \varepsilon < 0, \tag{A.451}$$

$$\varphi = \varphi_0 + D/r + \mathcal{O}\left(r^{-2}\right), \tag{452a}$$

$$g_{\mu\nu} = A^2(\varphi)\left[\tilde{g}_{\mu\nu} + \Lambda B^2(\varphi)\nabla_\mu\varphi\nabla_\nu\varphi\right], \tag{453}$$

$$\nabla_\mu F^{\mu\nu} = \nabla_\mu\nabla^\mu X^\nu - \nabla_\mu\nabla^\nu X^\mu. \tag{6.454}$$

$$\nabla_\mu\left(\hat{z}X^\mu\right) = 0, \tag{A.455}$$

$$V(\varphi) = \tfrac{1}{2}\mu^2\varphi^2 + \tfrac{1}{4}\lambda\varphi^4, \tag{456a}$$

$$\Box_\eta\,\varphi - \mu^2\varphi - \lambda\varphi^3 = 0, \tag{457}$$

$$\Box_\eta\,\delta\varphi - \mu^2\delta\varphi = 0. \tag{3.458}$$

$$\Box\,\delta\varphi - \mu^2\delta\varphi = 0, \tag{A.459}$$

$$\delta\varphi = \sum_{\ell m}\frac{\psi_{\ell m}(r)}{r}Y_{\ell m}(\theta,\phi)e^{-i\omega t}, \tag{460a}$$

$$\mathscr{G} = R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma} - 4R^{\mu\nu}R_{\mu\nu} + R^2, \tag{461}$$

$$g_{\mu\nu} \to C(\varphi)\left[g_{\mu\nu} + D(\varphi)\nabla_\mu\varphi\nabla_\nu\varphi\right] \tag{7.462}$$

$$\Box\varphi = -4\pi G_*\alpha(\varphi)T, \tag{A.463}$$

$$R_{\mu\nu} = 2\nabla_\mu\varphi\nabla_\nu\varphi + 8\pi G_*(T_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}\,T) \tag{464a}$$

$$g^{tt}\partial_t^2\delta\varphi + g^{xx}\partial_x^2\delta\varphi = \mu^2\varphi, \tag{465}$$

$$\omega(k) = \sqrt{(\mu^2 + g^{xx}k^2)/(-g^{tt})}. \tag{4.466}$$

$$X_\alpha \to X_\alpha + \mu_X^{-1}\nabla_\alpha\psi. \tag{A.467}$$

$$\overline{\Box}\psi = -\bar{g}^{\mu\nu}\left[\,\mu_X\overline{\nabla}_\mu X_\nu + \tfrac{1}{2}(\overline{\nabla}_\mu\log\hat{z})(\overline{\nabla}_\nu\psi + \mu_X X_\nu)\,\right], \tag{468a}$$

$$\nabla_\mu\nabla^\mu X_\nu + (\nabla_\mu\ln\hat{z})\nabla_\nu X^\mu = \mathcal{M}_{\nu\mu}X^\mu, \tag{469}$$

$$\mathcal{M}_{\nu\mu} = \hat{z}\mu_X^2 g_{\nu\mu} + R_{\nu\mu} - \nabla_\nu\nabla_\mu\ln|\hat{z}|. \tag{1.470}$$

$$\nabla_\mu\left(\hat{z}\nabla^\mu\varphi\right) = \mu^2\varphi, \tag{A.471}$$

$$\hat{z}\,\Box\delta\varphi + \cdots = 0. \tag{472a}$$

$$\mathcal{L}_\psi = \tfrac{1}{2}\left[\bar{\psi}\gamma^\mu(\nabla_\mu\psi) - \nabla_\mu\hat{\gamma}^5\gamma^\mu\psi\right] - \mu\bar{\psi}\psi, \tag{473}$$

$$\mathcal{L}_\psi^5 = \tfrac{1}{2}\left[\bar{\psi}\hat{\gamma}^5\gamma^\mu(\nabla_\mu\psi) - (\nabla_\mu\bar{\psi})\hat{\gamma}^5\gamma^\mu\psi\right] - \mu\bar{\psi}\psi, \tag{5.474}$$

$$\gamma^\mu\nabla_\mu\psi - \hat{\gamma}^5\mu\psi = 0. \tag{A.475}$$

$$\frac{N_I}{N_I+N_{II}} = \frac{1}{2}\left(1+V\cos\Delta\phi\right)\ , \tag{476a}$$

$$n(\mathbf{r}) = n_0 + V^{-1/2}\sum_{\mathbf{q}} n_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}}\ . \tag{477}$$

$$\sum_{f,\mathbf{p}'} |\langle \mathbf{p}', f|H_I|\mathbf{p}, i\rangle|^2 = \frac{1}{V}\int \frac{d^3\mathbf{q}}{(2\pi)^3}\frac{1}{16}\alpha_N^2\omega\,\omega' S(q), \tag{2.478}$$

$$\psi_{i,f} = \frac{1}{\sqrt{V}}e^{i\mathbf{p}\cdot\mathbf{r}}, \frac{1}{\sqrt{V}}e^{i\mathbf{p}'\cdot\mathbf{r}}\ . \tag{A.479}$$

$$t = 2M^2(1-\sqrt{1+q^2/M^2}) \quad \text{and} \quad s = m^2+M^2+2\omega M\ . \tag{480a}$$

$$\tilde{U}(\mathbf{q}) = \frac{e}{4\pi q^2}\left(Z+F(\mathbf{q})\right) \quad \text{where} \quad F(\mathbf{q}) = \int d^3\mathbf{r}\, n(\mathbf{r})\, e^{i\mathbf{q}\mathbf{r}}\ , \tag{481}$$

$$\mathbf{q} = (q\sin\theta_{\mathbf{qp}}\cos\phi_{\mathbf{qp}}, q\sin\theta_{\mathbf{qp}}\sin\phi_{\mathbf{qp}}, q\cos\theta_{\mathbf{qp}})\ , \tag{6.482}$$

$$s = (p+k)^2 = M^2+m^2+2\omega M, \qquad \text{and} \qquad t = (k-k')^2 = 2M^2\left(1-\sqrt{1+\frac{q^2}{M^2}}\right)\ . \tag{A.483}$$

$$d\cos\theta_{\mathbf{qp}} = -\frac{\omega}{q\sqrt{\omega^2-m^2}}d\omega\ , \tag{484a}$$

$$\frac{d\Phi}{d\omega} = \frac{dn}{d\omega}v = \frac{dn}{d\omega}\frac{\sqrt{\omega^2-m^2}}{\omega}\ . \tag{485}$$

$$f(\omega,\Omega) \equiv \omega\frac{dn}{d\omega d\Omega}\ . \tag{3.486}$$

$$d\theta_{\mathbf{p}}d\phi_{\mathbf{p}} = |J|\, d\alpha_{\mathbf{pq}}d\gamma_{\mathbf{pq}}\ , \tag{A.487}$$

$$f(\omega,\Omega_{\mathbf{p}},t) = \omega\frac{dn(\omega,\Omega_{\mathbf{p}})}{d\omega d\Omega_{\mathbf{p}}} \to \frac{\omega}{4\pi}\frac{dn(\omega)}{d\omega}\ . \tag{488a}$$

$$z_a(t|\hat{\Omega}) = \frac{1}{2}\frac{n_a^i n_a^j}{1+\hat{\Omega}\cdot\hat{n}_a}\Delta h_{ij}\,, \tag{489}$$

$$F_a^A(\hat{\Omega}) = \frac{1}{2}\frac{n_a^i n_a^j e_{ij}^A(\hat{\Omega})}{(1+\hat{\Omega}\cdot\hat{n}_a)}\,. \tag{7.490}$$

$$z(f,\hat{n}) = \sum_{\ell=2}^{\infty}\sum_{m=-\ell}^{\ell} z_{\ell m}(f)Y_{\ell m}(\hat{n})\,, \tag{A.491}$$

$$\langle \tilde{h}_A^*(f,\hat{\Omega})\tilde{h}_{A'}(f',\hat{\Omega}')\rangle = \delta(f-f')\delta^2(\hat{\Omega},\hat{\Omega}')\mathcal{P}_{A,A'}(f,\hat{\Omega})\,. \tag{492a}$$

$$\mathcal{P}_{A,A'}(f,\hat{\Omega}) = \begin{pmatrix} I(f,\hat{\Omega})+Q(f,\hat{\Omega}) & U(f,\hat{\Omega})-iV(f,\hat{\Omega}) \\ U(f,\hat{\Omega})+iV(f,\hat{\Omega}) & I(f,\hat{\Omega})-Q(f,\hat{\Omega}) \end{pmatrix}\ . \tag{493}$$

$$c_{LM}^{+} = (-1)^{M} (c_{L,-M}^{-})^{*} \,, \tag{4.494}$$

$$c_{L0\ +4}^{X} Y_{LM}(\hat{\Omega}) \leq \frac{1}{\sqrt{4\pi}} \,, \tag{A.495}$$

$$\left\langle \left| \widehat{A_{\ell\ell'}^{LM}} \right|^{2} \right\rangle = \left[1 + (-1)^{\ell+\ell'+L} \delta_{\ell\ell'} \right] C_{\ell} C_{\ell'} . \tag{496a}$$

$$z(\hat{n}|\hat{z}) = \frac{1}{2} \left[\tilde{h}_{+} (1 - \cos\theta) \cos 2\phi + \tilde{h}_{\times} (1 - \cos\theta) \sin 2\phi \right] \,, \tag{497}$$

$$H_{\ell\ell'}^{L} \equiv (-1)^{L} \begin{pmatrix} \ell & \ell' & L \\ 2 & 2 & -4 \end{pmatrix} . \tag{1.498}$$

$$r \equiv \frac{\left\langle \widehat{c_{LM}^{E}} \widehat{c_{LM}^{I}} \right\rangle}{\left(\Delta c_{LM}^{E} \right) \left(\Delta c_{LM}^{I} \right)} \,. \tag{A.499}$$

$$\left(\widehat{c_{LM}^{V}} \right)_{\ell,\ell'} = \frac{1}{\sqrt{4\pi} z_{\ell} z_{\ell'}} \frac{\widehat{A_{\ell\ell'}^{LM}}}{G_{\ell\ell'}^{L} I(f)} \,, \tag{500a}$$

$$\left\langle \left| \widehat{A_{\ell\ell'}^{LM}} \right|^{2} \right\rangle = C_{\ell} C_{\ell'} \,. \tag{501}$$

$${}^{(ab)}\Gamma_{LM}^{X} = \sum_{\ell\ell'} z_{\ell} z_{\ell'} \sqrt{4\pi} H_{\ell\ell'}^{L} \left\{ Y_{\ell}(\hat{n}_{a}) \otimes Y_{\ell'}(\hat{n}_{b}) \right\}_{LM} \,, \tag{5.502}$$

$$\kappa_{ab}(f, \hat{\Omega}) \equiv \left[1 - e^{-2\pi i f L_{a} (1 + \hat{\Omega} \cdot \hat{p}_{a})} \right] \left[1 - e^{2\pi i f L_{b} (1 + \hat{\Omega} \cdot \hat{p}_{b})} \right] . \tag{A.503}$$

$$c_{LM}^{E} = \frac{1}{2} \left(c_{LM}^{+} + c_{LM}^{-} \right) , \quad c_{LM}^{B} = -\frac{i}{2} \left(c_{LM}^{+} - c_{LM}^{-} \right) \,. \tag{504a}$$

$$\begin{aligned} \langle z(\hat{n}_{a}) z(\hat{n}_{b}) \rangle &= I(f) \zeta(\theta_{ab}) \\ &\quad + I(f) \sum_{LM} \left[c_{LM}^{E}\, {}^{(ab)}\Gamma_{LM}^{E} + i c_{LM}^{B}\, {}^{(ab)}\Gamma_{LM}^{B} \right] , \end{aligned} \tag{505}$$

$$\pi(G, C_{t'}) \;=\; \lambda_{t'}(E(G)) \,. \tag{2.506}$$

$$\mathbb{E} \left(\lambda_{t+\tau+\phi}(F^{S}) \right) \;\leq\; (1 - \beta) \cdot \left(\lambda_{t+\tau+\phi-1}(F^{S}) \right) \,, \tag{A.507}$$

$$\mathbb{E} \left(\lambda_{t+\tau+\phi}^{\sigma}(S) + \lambda_{t+\tau+\phi}^{\rho}(F^{S}) \right) \;\leq\; (1 - \beta) \cdot \left(\lambda_{t+\tau+\phi-1}^{\sigma}(S) + \lambda_{t+\tau+\phi-1}^{\rho}(F^{S}) \right) \,, \tag{508a}$$

$$\mathbb{P} \left(A_{e} \right) \geq \frac{1}{4\phi^{2}} . \tag{509}$$

$$\mathbb{P} \left(A_{v} \right) \geq 2^{-\phi} . \tag{6.510}$$

$$\mathbb{P} \left(Y < 3k \right) = \mathbb{P} \left(Y < (1 - 1/4) \cdot \xi k 2^{-(\phi+1)} \right) \leq \mathbb{P} \left(Y < (1 - 1/4) \cdot \mathbb{E} \left(Y \right) \right) < e^{-k/8} \leq e^{-1/8} \,. \tag{A.511}$$

$$P_{\phi}^{t}(j_0, j) \geq \boldsymbol{\pi}(j) \cdot (1 - \epsilon). \tag{512a}$$

$$\pi(G, C) \;=\; \sum_{e \in \overline{D}(C)} \sigma(C[e]) \tag{513}$$

$$O\left(\left(\phi^5/\beta\right) \cdot (\log(k) + (\psi + \phi)\log(\Delta) + \log(\sigma_0))\right). \tag{3.514}$$

$$\mathbb{E}(\pi(G, C_{t+\tau+\phi})) \;\leq\; \left(1 - \frac{\beta}{4\phi^2}\right) \cdot \pi(G, C_t)\,. \tag{A.515}$$

$$S_t^j \;=\; \left\{v \in V(G_t) \mid \texttt{step}_{v,t} = j\right\}. \tag{516a}$$

$$Q \;=\; \left\{\{u, v\} \in E(H) \cap \overline{D}(C_t) \mid u, v \in S_t^{\phi-1}\right\} \quad \text{and} \quad R \;=\; \{\texttt{out}_u(v) \mid \{u, v\} \in A \cap Q\}\,. \tag{517}$$

$$M^* \;\subseteq\; M_< \;\subseteq\; M_< + M_{\geq}^{D,t+1} + M_{\geq}^{\overline{D},t+1} \;\subseteq\; M_< + M_{\geq}^{D,t} + M_{\geq}^{\overline{D},t+1}\,. \tag{7.518}$$

$$Q \;=\; \left\{\{u, v\} \in \overline{D}(C_t) \mid u, v \in S_t^{\phi-1}\right\} \quad \text{and} \quad R \;=\; \{\texttt{out}_u(v) \mid \{u, v\} \in Q\}\,. \tag{A.519}$$

$$\mathbb{E}\left(\pi(H_{t+\tau+\phi}, C_{t+\tau+\phi}^H)\right) \;\leq\; (1 - \beta) \cdot \pi(H_{t+\tau+\phi-1}, C_{t+\tau+\phi-1}^H)\,, \tag{520a}$$

$$O\left(\left(\phi^5/\beta\right) \cdot (\log(k) + (\psi + \phi)\log(\Delta) + \log(\sigma_0 + \rho_0))\right). \tag{521}$$

$$\mathbb{E}\left(\min\{i \geq 0 \mid P_i = 0\}\right) \;\leq\; O\left(\left(\phi^2/\beta\right) \cdot (\log(\sigma_0 + \rho_0) + \log(k) + (\psi + \phi)\log(\Delta))\right). \tag{4.522}$$

$$M_< = \{C_{t^\circ}(u)\}_{u \in A_<}\,. \tag{A.523}$$

$$Q \;=\; \overline{D}(C_t) \cap S_t^{\phi-1} \quad \text{and} \quad R \;=\; \{\texttt{out}_v \mid v \in Q\}\,. \tag{524a}$$

$$\sigma(M) \;=\; 3c - \sum_{i=1}^{c-1} \min\{M(i), 2\}\,. \tag{525}$$

$$|\{u \in N_H(v) \,:\, \mathrm{d}_H(u) \leq \mathrm{d}_H(v)\}| \;\geq\; \mathrm{d}_H(v)/3\,. \tag{1.526}$$

$$\lambda'(F_v) \;<\; \lambda(F_v) - c \cdot |F_v^{g,c}| \;<\; \lambda(F_v) - \frac{c}{4}|F_v^g| \;\leq\; \lambda(F_v) - \frac{c}{12}|F_v| \;\leq\; \lambda(F_v) - \frac{1}{36}\lambda(F_v) \;=\; \frac{35}{36}\lambda(F_v)\,, \tag{A.527}$$

$$X_e \;=\; \begin{cases} 2^{\mathrm{ld}(u)}\,, & e \in R \\ 0\,, & \text{otherwise} \end{cases} \tag{528a}$$

$$\sigma(M) \;=\; 1\,. \tag{529}$$

$$\sigma(M) = 1 \quad \text{and} \quad c \;\leq\; \rho(M) \;\leq\; 2c \tag{5.530}$$

$$\lambda'(\{u, v\}) \;=\; \begin{cases} \rho(C'[u]) + \rho(C'[v])\,, & u, v \in \overline{D}(C') \\ 0\,, & \text{otherwise} \end{cases} \tag{A.531}$$

$$|U^0| + \lambda(E(H)) \geq \frac{c}{c+1}\pi(G, C)\,. \tag{532a}$$

$$\sum_{i=1}^{c-1} C[v](i) \geq c - 1\,. \tag{533}$$

$$\lambda'(F_v) \leq \lambda(F_v) - 2c|U_v^{g,c}| < \lambda(F_v) - c|U_v^g| \leq \lambda(F_v) - \frac{c}{3}|F_v| \leq \lambda(F_v) - \frac{1}{12}\lambda(F_v) = \frac{11}{12}\lambda(F_v)\,, \tag{2.534}$$

$$\lambda'(F_v) \leq \frac{4c-1}{4c}\lambda(F_v)\,, \tag{A.535}$$

$$x_i^{(k+1)} = (1-\omega)x_k^{(k)} + \frac{\omega}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\Big), \quad i = 1, \ldots, m, \tag{536a}$$

$$x_i^{(k+1)} = \max\Big[(1-\omega)x_k^{(k)} + \frac{\omega}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\Big), 0\Big], \quad i = 1, \ldots, m. \tag{537}$$

$$\begin{aligned} \widetilde{\boldsymbol{x}}^{(k+1)} &= G_{\mathrm{SOR}}\boldsymbol{x}^{(k)} + \boldsymbol{c}_{\mathrm{SOR}}, \\ \boldsymbol{x}^{(k+1)} &= \max\{\widetilde{\boldsymbol{x}}^{(k+1)}, \boldsymbol{0}\}, \end{aligned} \tag{6.538}$$

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - hP\nabla_{\mathrm{d}}V(\boldsymbol{x}^{(k+1)}, \boldsymbol{x}^{(k)}) \tag{A.539}$$

$$V(\boldsymbol{x}^{(k+1)}) \leq V(\boldsymbol{x}^{(k)}), \tag{540a}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) = -P\nabla V(\boldsymbol{x}(t)). \tag{541}$$

$$V(\boldsymbol{x}^{(k+1)}) - V(\boldsymbol{x}^{(k)}) \leq -\gamma\|\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^{(k)}\|^2 \leq 0, \tag{3.542}$$

$$\gamma = \frac{1}{h}\min(a_{11}, a_{22}, \ldots, a_{nn}). \tag{A.543}$$

$$h^{(k)} = \frac{2\omega^{(k)}}{2 - \omega^{(k)}}. \tag{544a}$$

$$V(\boldsymbol{x}^{(k+1)}) - V(\boldsymbol{x}^{(k)}) \leq -\tilde{\gamma}\|\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^{(k)}\|^2 \leq 0, \tag{545}$$

$$\tilde{\gamma} = \frac{1}{M_h}\min(a_{11}, a_{22}, \ldots, a_{nn}). \tag{7.546}$$

$$\overline{s}_m^{(k')} > \overline{s}_m^{(k'-1)} \tag{A.547}$$

$$C \leq V\big(\boldsymbol{x}^{(k+1)}\big) \leq V\big(\boldsymbol{x}^{(k)}\big) \leq \cdots \leq V\big(\boldsymbol{x}^{(0)}\big), \tag{548a}$$

$$\lim_{k\to\infty} V\big(\boldsymbol{x}^{(k)}\big) = V^*. \tag{549}$$

$$V_t = \{\boldsymbol{x} \in \mathbb{R}^m \mid V(\boldsymbol{x}) \leq t\} \tag{4.550}$$

$$\lim_{l\to\infty} V\big(\boldsymbol{x}^{(k_l)}\big) = V(\overline{\boldsymbol{x}}). \tag{A.551}$$

$$0 = \lim_{l\to\infty}\Big(V\big(\boldsymbol{x}^{(k_l+1)}\big) - V\big(\boldsymbol{x}^{(k_l)}\big)\Big) \le -\lim_{l\to\infty}\gamma\big\|\boldsymbol{x}^{(k_l+1)} - \boldsymbol{x}^{(k_l)}\big\|^2 \le 0. \tag{552a}$$

$$\lim_{l\to\infty}\boldsymbol{x}^{(k_l+1)} = \lim_{l\to\infty}\boldsymbol{x}^{(k_l)} = \overline{\boldsymbol{x}}. \tag{553}$$

$$\alpha = \frac{h}{1+\frac{h}{2}}, \quad E = D^{-1}, \tag{1.554}$$

$$d^{(k)} := \log_{10}\|\boldsymbol{x}^{(k)} - \boldsymbol{x}^{(k-1)}\|. \tag{A.555}$$

$$\overline{s}_m^{(k)} = \frac{1}{m}\big(s^{(k)} + s^{(k-1)} + \cdots + s^{(k-m+1)}\big) = \frac{1}{m}\big(d^{(k)} - d^{(k-m)}\big), \tag{556a}$$

$$\overline{h}_m^{(k)} = \frac{1}{m+1}\big(h^{(k-1)} + h^{(k-2)} + \cdots + h^{(k-m-1)}\big) \tag{557}$$

$$\overline{\omega}_m^{(k)} = \frac{1}{m+1}\big(\omega^{(k-1)} + \omega^{(k-2)} + \cdots + \omega^{(k-m-1)}\big). \tag{5.558}$$

$$h^{(k)} = \overline{h}_m^{(k')} \quad \text{or equivalently} \quad \omega^{(k)} = \overline{\omega}_m^{(k')} \tag{A.559}$$

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \tag{560a}$$

$$A = \begin{bmatrix} 2 & -1 & 0.5 \\ -1 & 2 & -1 \\ 0.5 & -1 & 2 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}, \tag{561}$$

$$x_i^{(k+1)} = x_i^{(k)} - \frac{h}{a_{ii}}\bigg(\sum_{j<i} a_{ij}x_j^{(k+1)} + a_{ii}\frac{x_i^{(k+1)} + x_i^{(k)}}{2} + \sum_{j>i} a_{ij}x_j^{(k)} - b_i\bigg). \tag{2.562}$$

$$x_i^{(k+1)} = \frac{-1}{a_{ii}\big(1+\frac{h}{2}\big)}\bigg[h\sum_{j<i} a_{ij}x_j^{(k+1)} - \bigg(1-\frac{h}{2}\bigg)a_{ii}x_i^{(k)} + h\sum_{j>i} a_{ij}x_j^{(k)} - hb_i\bigg]. \tag{A.563}$$

$$\begin{aligned} &V(x_1^{(k+1)}, \ldots, \hat{x}_i^{(k+1)}, x_{i+1}^{(k)}, \ldots, x_n^{(k)}) - V(x_1^{(k+1)}, \ldots, x_{i-1}^{(k+1)}, x_i^{(k)}, \ldots, x_n^{(k)}) \\ &\quad = -\frac{a_{ii}}{h}(\hat{x}_i^{(k+1)} - x_i^{(k)})^2 \le 0. \end{aligned} \tag{564a}$$

$$\begin{aligned} &V(x_1^{(k+1)}, \ldots, x_i^{(k+1)}, x_{i+1}^{(k)}, \ldots, x_n^{(k)}) - V(x_1^{(k+1)}, \ldots, x_{i-1}^{(k+1)}, x_i^{(k)}, \ldots, x_n^{(k)}) \\ &\quad \le -\frac{a_{ii}}{h}(x_i^{(k+1)} - x_i^{(k)})^2 \le 0 \end{aligned} \tag{565}$$

$$p_k(x; \mathbf{X}) := \frac{k-1}{U_k(x; \mathbf{X})}, \tag{6.566}$$

$$\tilde{p}_{k,M}^{\mathsf{AM}}(x; \mathbf{X}_{1:M}) := \frac{1}{M}\sum_{m=1}^{M} \tilde{p}_k(x; \mathbf{X}_m). \tag{A.567}$$

$$\mathbb{E}_{\mathbf{X}_{1:M}}[(\tilde{p}^{\mathsf{AM}}_{k,M}(x;\mathbf{X}_{1:M})-p(x))^2]=\tilde{O}(n^{-2\zeta}+M^{-1}). \tag{568a}$$

$$\mathbb{E}[(p^{\mathsf{AM}}_{k,M}-p(x))^2]=\tilde{O}(N^{-\frac{2\zeta}{1+2\zeta}}). \tag{569}$$

$$p^{\mathsf{HM}}_{k,M}(x):=p^{\mathsf{HM}}_{k,M}(x;\mathbf{X}_{1:M}):=\frac{kM}{\sum_{m=1}^{M}U_k(x;\mathbf{X}_m)}. \tag{3.570}$$

$$p^{\mathsf{GM}}_{k,M}(x;\mathbf{X}_{1:M}):=e^{\Psi(k)}\Big(\prod_{m=1}^{M}\frac{1}{U_k(x;\mathbf{X}_m)}\Big)^{\frac{1}{M}}, \tag{A.571}$$

$$f_\alpha(p):=\begin{cases}\frac{p^\alpha}{\alpha} & \text{if } \alpha\neq 0,\\ \log p & \text{if } \alpha=0.\end{cases} \tag{572a}$$

$$\eta_{k,M}(x):=\eta_{k,M}(x;\mathscr{P}):=\frac{1}{M}\sum_{m=1}^{M}\eta_k(x;\mathcal{D}_m). \tag{573}$$

$$\mathbb{E}_{\mathscr{P}}\|\eta_{k,M}-\eta\|_2\le C_1\Big\{\Big(\frac{M\log M}{N}\Big)^{\frac{\alpha_{\mathrm{H}}}{d}}+\sqrt{\frac{(\log M)^{1+\epsilon}}{M}}\Big\}. \tag{7.574}$$

$$\mathscr{P}(g_{k,M}(x)\neq g^*(X)|\mathscr{P})\le\delta+C_o\Big(\frac{\log N}{N}\log\frac{2}{\delta}\Big)^{\frac{\alpha\beta}{2\alpha+1}}. \tag{A.575}$$

$$\mathbb{E}_{\mathscr{P}}\|\eta_{k,M,L}-\eta\|_2=O\big(N^{-\frac{\alpha_{\mathrm{H}}}{2\alpha_{\mathrm{H}}+d}}\big). \tag{576a}$$

$$T_{M;S}(k,N)=\Big\lceil\frac{M}{S}\Big\rceil T_{\mathrm{NN}}\Big(k,\Big\lceil\frac{N}{M}\Big\rceil\Big). \tag{577}$$

$$\frac{T_{M;S}(k,N)}{T_{\mathrm{NN}}(kM,N)}\sim\frac{1}{S} \tag{4.578}$$

$$\frac{T_{M;S}(k,N)}{T_{\mathrm{NN}}(kM,N)}\sim\frac{\frac{kM}{S}\log\frac{N}{M}}{kM\log N}=\frac{1}{S}\Big(1-\frac{\log M}{\log N}\Big) \tag{A.579}$$

$$\eta_k(x;\mathscr{D}):=\frac{1}{k}\sum_{i=1}^{k}Y_{(i)}(x;\mathscr{D}). \tag{580a}$$

$$\mu(\mathbb{B}^o(x,r))\le 2^d\mu\Big(\mathbb{B}^o\big(x,\frac{r}{2}\big)\Big). \tag{581}$$

$$\mu(\mathbb{B}^o(x,r))\ge\min\{C_d r^d,1\}. \tag{1.582}$$

$$|\eta(x)-\eta(x')|\le A\rho^{\alpha_{\mathrm{H}}}(x,x'). \tag{A.583}$$

$$R(\hat{g})-R^*\le 2\|\hat{\eta}-\eta\|_1, \tag{584a}$$

$$\eta_{k,M,L}(x):=\eta_{k,M,L}(x;\mathscr{P}):=\frac{1}{L}\sum_{j=1}^{L}\eta_k(x;\mathcal{D}_{m_j}). \tag{585}$$

$$p_{k,M}^{\mathsf{HM}}(x;\mathbf{X}_{1:M}) = \frac{kM-1}{(k-1)M}\Big(\frac{1}{M}\sum_{m=1}^{M} p_k(x;\mathbf{X}_m)^{-1}\Big)^{-1}, \tag{5.586}$$

$$p_{k,M}^{(f)}(x;\mathbf{X}_{1:M}) := f^{-1}\Big(\frac{1}{M}\sum_{m=1}^{M}\phi_k^{(f)}(U_k(x;\mathbf{X}_m))\Big). \tag{A.587}$$

$$\phi_k^{(f)}(u) = \frac{\Gamma(k)}{u^{k-1}}\mathcal{L}^{-1}\Big\{\frac{f(p)}{p^k}\Big\}(u) \quad \text{for } u>0, \tag{588a}$$

$$\boldsymbol{t}_i'(\boldsymbol{n}_i,\boldsymbol{t}_i|\boldsymbol{\gamma}) = \boldsymbol{t}_i\sum_{b=1}^{B^2}\gamma_b\Phi_b(\boldsymbol{n}_i) \tag{589}$$

$$\boldsymbol{I}(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\delta},\boldsymbol{\gamma},\boldsymbol{p}) = \Pi(\boldsymbol{RF}(\boldsymbol{S},\boldsymbol{T}')+\boldsymbol{t}) \tag{2.590}$$

$$L_{pho}(\boldsymbol{x}) = \frac{\sum_{i\in\boldsymbol{\mathcal{M}}}\boldsymbol{A}_i\cdot\left\|\boldsymbol{I}_i-\boldsymbol{I}_i'(\boldsymbol{x})\right\|_2}{\sum_{i\in\boldsymbol{\mathcal{M}}}\boldsymbol{A}_i} \tag{A.591}$$

$$L_{lan}(\boldsymbol{x}) = \frac{1}{N}\sum_{n=1}^{N}\omega_n\left\|l_n-l_n'(\boldsymbol{x})\right\|_2 \tag{592a}$$

$$L_{reg}(\boldsymbol{x}) = \omega_{\boldsymbol{\alpha}}\left\|\boldsymbol{\alpha}\right\|^2+\omega_{\boldsymbol{\beta}}\left\|\boldsymbol{\beta}\right\|^2+\omega_{\boldsymbol{\delta}}\left\|\boldsymbol{\delta}\right\|^2 \tag{593}$$

$$L_{regTex}(\boldsymbol{x}) = \frac{1}{T}\sum_{i\in\boldsymbol{\mathcal{T}}}\left\|\boldsymbol{t}_i(\boldsymbol{\delta})-\mu\right\|^2 \tag{6.594}$$

$$L(\boldsymbol{x}) = \lambda_{pho}L_{pho}(\boldsymbol{x})+\lambda_{lan}L_{lan}(\boldsymbol{x})+\lambda_{reg}L_{reg}(\boldsymbol{x}) \tag{A.595}$$

$$\boldsymbol{Z}_{\boldsymbol{S}}' = \frac{\boldsymbol{Z}_{\boldsymbol{S}}-\boldsymbol{Z}_{min}}{\boldsymbol{Z}_{max}-\boldsymbol{Z}_{min}}, \tag{596a}$$

$$\rho\geq 0, \quad \operatorname{Tr}\rho = 1. \tag{597}$$

$$\rho_{\mathcal{A}} = \operatorname{Tr}_{\mathcal{B}}\rho, \tag{3.598}$$

$$\rho = \sum_i p_i\rho_{\mathcal{A},i}\otimes\rho_{\mathcal{B},i}, \tag{A.599}$$

$$\Pi_x\geq 0, \quad \sum_{x\in\mathcal{X}}\Pi_x = I_{\mathcal{H}}. \tag{600a}$$

$$\begin{aligned}\operatorname{Tr}\kappa(\rho) &= 1 \quad \forall\rho\in\mathcal{S}(\mathcal{H}),\\ \kappa\otimes\iota_{\mathbb{C}^d}(\rho) &\geq 0 \quad \forall\rho\in\mathcal{S}(\mathcal{H}\otimes\mathbb{C}^d),\ \forall d\geq 1,\end{aligned} \tag{601}$$

$$I(A;B)[\rho_{AB}] := D(\rho_{AB}\|\rho_A\otimes\rho_B), \tag{7.602}$$

$$I(M;\mathcal{B}) = 0 \tag{A.603}$$

$$\Gamma[m](\rho) := \sum_{u_D \in \mathbb{F}_q^{\mathsf{y}}} \frac{1}{q^{\mathsf{y}_2}} \mathbf{W}_{[\mathsf{n}]}(Fm + G^{(2)}u_D)\rho \mathbf{W}_{[\mathsf{n}]}^{\dagger}(Fm + G^{(2)}u_D). \tag{604a}$$

$$\sum_{a \in \mathbf{F}_q^{\mathsf{y}'-1}} \left| F' \begin{pmatrix} 0 \\ a \end{pmatrix} \right\rangle. \tag{605}$$

$$N = \{G^{(1)}y\}_{y \in \mathbb{F}_q^{(\mathsf{n}-\mathsf{x})}}, \quad N^{\perp} = \{G^{(1)}y + Fx\}_{y \in \mathbb{F}_q^{\mathsf{y}_1}, x \in \mathbb{F}_q^{2\mathsf{x}}}. \tag{4.606}$$

$$\Gamma[m](\rho) := \sum_{u_D \in \mathbb{F}_q^{\mathsf{y}}} \frac{1}{q^{\mathsf{y}}} \mathbf{W}_{[\mathsf{n}]}(Fm + Gu_D)\rho \mathbf{W}_{[\mathsf{n}]}^{\dagger}(Fm + Gu_D). \tag{A.607}$$

$$\Gamma[m](\rho) := \sum_{u_{D,2} \in \mathbb{F}_q^{\mathsf{y}_2}} \frac{1}{q^{\mathsf{y}_2}} \mathbf{W}_{[\mathsf{n}]}(Fm + G^{(2)}u_{D,2})\rho \mathbf{W}_{[\mathsf{n}]}^{\dagger}(Fm + G^{(2)}u_{D,2}). \tag{608a}$$

$$\Gamma[m](\rho) := \sum_{u_D \in \mathbb{F}_q^{\mathsf{y}}} \frac{1}{q^{\mathsf{y}}} \mathbf{W}_{[\mathsf{n}]}(Fm + G^{(2)}U_{D,2})\rho \mathbf{W}_{[\mathsf{n}]}^{\dagger}(Fm + G^{(2)}U_{D,2}). \tag{609}$$

$$\operatorname{Tr} \Lambda(X)Y = \operatorname{Tr} X\Lambda^*(Y). \tag{1.610}$$

$$I(X; BR)[\rho_{BRX}] = I(R; B)[\sigma_{RB}]. \tag{A.611}$$

$$\Lambda(\rho) := \operatorname{Tr}_{\mathcal{A}^c} \sum_{u_{D,2} \in \mathbb{F}_q^{\mathsf{y}_2}} \frac{1}{q^{\mathsf{y}_2}} \mathbf{W}_{[\mathsf{n}]}(G^{(2)}u_{D,2})\rho \mathbf{W}_{[\mathsf{n}]}^{\dagger}(G^{(2)}u_{D,2}). \tag{612a}$$

$$P_{\overline{\{1\}}}G^{(1)} = P_{\overline{\{2\}}}G^{(1)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_{\overline{\{3\}}}G^{(1)} = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}, \tag{613}$$

$$P_{\overline{\{1,3\}}}G^{(1),*} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 2 \\ 0 & 1 & 0 \\ 0 & 2 & 2 \end{pmatrix}, \quad P_{\overline{\{1,3\}}}F^* = \begin{pmatrix} 2 & 0 \\ 1 & 2 \\ 0 & 0 \\ 0 & 2 \end{pmatrix}, \tag{5.614}$$

$$\sum_{i=0}^{n} (\sum_{j=1}^{\mathsf{d}} \beta_{j,i} a_{\mathsf{d}+1,j}) \alpha^i = \sum_{j=1}^{\mathsf{d}} \sum_{i=0}^{n} \beta_{j,i} \alpha^i a_{\mathsf{d}+1,j} = a_{\mathsf{d}+1,\mathsf{d}+1} = \alpha. \tag{A.615}$$

$$\sum_{j=1}^{\mathsf{d}} \beta_{j,1} a_{\mathsf{d}+1,j} = 1, \tag{616a}$$

$$\sum_{j=1}^{\mathsf{d}} \beta_{j,1} b^j = 0. \tag{617}$$

$$A = \begin{pmatrix} A_2 \\ A_3 \\ A_1 \\ I \end{pmatrix}, \quad B = \begin{pmatrix} I & 0 & 0 \\ -A_1^T & A_2^T & A_3^T \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix}. \tag{2.618}$$

$$C = \begin{pmatrix} C_2 \\ C_3 \\ C_1 \\ 0 \end{pmatrix}. \tag{A.619}$$

$$J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}. \tag{620a}$$

$$\operatorname{Tr} \rho[m](Y_{\mathcal{A}}(m) \otimes I_{\mathcal{A}^c}) = 1 \tag{621}$$

$$\mathcal{DECC}[\mathcal{A}](\operatorname{Tr}_{\mathcal{A}^c} \Gamma(\rho)) = \rho \tag{6.622}$$

$$\mathsf{Enc}_{\text{user}}(K) = Q^{(K)} = (Q_1^{(K)}, \ldots, Q_{\mathsf{n}}^{(K)})^T \in \mathcal{Q}_1 \times \cdots \times \mathcal{Q}_{\mathsf{n}}, \tag{A.623}$$

$$\mathsf{Enc}_{\text{serv}_j}(\vec{M}, Q_j^{(K)}, R_j) = \Lambda_j. \tag{624a}$$

$$\operatorname{Tr} \rho(\vec{m}, q, k)(Y_{k,q,\mathcal{A}}(m_k) \otimes I_{\mathcal{A}^c}) = 1 \tag{625}$$

$$\operatorname{Tr} \Gamma[m](\rho_{DE})(Y_{\mathcal{A},E}(m) \otimes I_{\mathcal{A}^c}) = 1 \tag{3.626}$$

$$\min_{\boldsymbol{\theta}} \ F(\boldsymbol{\theta}) + \frac{\rho}{2} \operatorname{dist}(\boldsymbol{\theta}, C)^2, \tag{A.627}$$

$$\min_{\boldsymbol{\theta}} \ F(\boldsymbol{\theta}) + \frac{\rho}{2} \|\boldsymbol{\theta} - P_C(\boldsymbol{\theta})\|^2. \tag{628a}$$

$$\boldsymbol{\theta}_k = P_C(\boldsymbol{\theta}_{k-1}) - \frac{1}{\rho_k} \nabla F(\boldsymbol{\theta}_k). \tag{629}$$

$$\begin{aligned} \boldsymbol{\eta}_k &= P_C[\boldsymbol{\theta}_k - \alpha_k \nabla f(\boldsymbol{\eta}_k; \boldsymbol{z}_{\xi_k})], \\ \boldsymbol{\theta}_{k+1} &= P_C[\boldsymbol{\eta}_k - \alpha_k \nabla g(\boldsymbol{\eta}_k; \boldsymbol{z}_{\xi_k})]. \end{aligned} \tag{7.630}$$

$$\sum_{k=1}^{\infty} \frac{2}{\rho_k} \{F[P_C(\boldsymbol{\theta}_{k-1})] - F(\boldsymbol{\theta}_*)\} \ < \ \infty. \tag{A.631}$$

$$\min_{\theta} \ \frac{1}{b} \sum_{i=1}^{b} L_\delta(\tilde{y}_i - \tilde{\boldsymbol{x}}_i^T \boldsymbol{\theta}) + \frac{\rho_k}{2} \|\boldsymbol{\theta} - P_C(\boldsymbol{\theta}_{k-1})\|^2 := G(\boldsymbol{\theta}) \tag{632a}$$

$$\mathbb{E}(\nabla f(\boldsymbol{\theta}_k; \boldsymbol{z}_{\xi_k}) | \mathcal{F}_{k-1}) = \frac{1}{n} \sum_{i=1}^{n} \nabla f(\boldsymbol{\theta}_k^i; \boldsymbol{z}_i) \neq \nabla F(\boldsymbol{\theta}_k^+). \tag{633}$$

$$\boldsymbol{\theta}_k = \operatorname{prox}_{\rho^{-1} F}[P_C(\boldsymbol{\theta}_{k-1})] = \operatorname{prox}_{\rho^{-1} F}[\boldsymbol{\theta}_{k-1} - \rho^{-1} \nabla q_\rho(\boldsymbol{\theta}_{k-1})], \tag{4.634}$$

$$\beta_j = \beta_{j-1} - \eta\{-\frac{1}{b} \tilde{X}^T \tilde{l} + \rho_k[\beta_{j-1} - P_C(\theta_{k-1})]\} \tag{A.635}$$

$$\log C_k = \sum_{i=1}^{k} \log(1 + a_i) \leq \sum_{i=1}^{k} a_i \leq \sum_{i=1}^{+\infty} a_i. \tag{636a}$$

$$\mathbb{E}\{\|\nabla f[P_C(\boldsymbol{\theta}_k); \boldsymbol{z}_{\xi_k}]\|^2\} \leq 2(G^2 + r^2L^2) := c^2. \tag{637}$$

$$\sum_{k=1}^{\infty} \frac{2}{\rho_k}\{F[P_C(\boldsymbol{\theta}_{k-1})] - F(\boldsymbol{\theta}_*)\} \ = \ \infty, \tag{1.638}$$

$$\frac{1}{\rho_k^2}\mathbb{E}\{\|\nabla f[P_C(\boldsymbol{\theta}_{k-1}); \boldsymbol{z}_{\xi_k}]\|^2\} \leq \frac{c^2}{\rho_k^2} \tag{A.639}$$

$$\min_{\boldsymbol{\theta}\in C} \ F(\boldsymbol{\theta}) = \frac{1}{2n}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i^T\boldsymbol{\theta})^2, \tag{640a}$$

$$\frac{1}{b}\sum_{i=1}^{b} -(\tilde{y}_i - \tilde{\boldsymbol{x}}_i^T\boldsymbol{\theta})\tilde{\boldsymbol{x}}_i = -\frac{1}{b}(\tilde{\boldsymbol{X}}^T\tilde{\boldsymbol{y}} - \tilde{\boldsymbol{X}}^T\tilde{\boldsymbol{X}}\boldsymbol{\theta}), \tag{641}$$

$$\min_{\boldsymbol{\theta}\in C} \ F(\boldsymbol{\theta}) = -\frac{1}{n}\sum_{i=1}^{n}\{y_i(\boldsymbol{x}_i^T\boldsymbol{\theta}) - log(1 + e^{\boldsymbol{x}_i^T\boldsymbol{\theta}})\}. \tag{5.642}$$

$$g_k(\boldsymbol{\theta}) = -\frac{1}{b}\sum_{i=1}^{b}\{\tilde{y}_i(\tilde{\boldsymbol{x}}_i^T\boldsymbol{\theta}) - log(1 + e^{\tilde{\boldsymbol{x}}_i^T\boldsymbol{\theta}})\}, \tag{A.643}$$

$$F(\boldsymbol{\theta}_3) < \lambda F(\boldsymbol{\theta}_1) + (1-\lambda)F(\boldsymbol{\theta}_2) = \min_{\boldsymbol{\theta}\in C} \ F(\boldsymbol{\theta}). \tag{644a}$$

$$\|\boldsymbol{x}_k - \boldsymbol{y}\|^2 \ \leq \ \|\boldsymbol{x}_{k-1} - \boldsymbol{y}\|^2 - \frac{2}{\rho_k}[g(\boldsymbol{x}_k) - g(\boldsymbol{y})] - \|\boldsymbol{x}_{k-1} - \boldsymbol{x}_k\|^2 \tag{645}$$

$$\mathbb{E}\{\nabla f[P_C(\boldsymbol{\theta}_{k-1}); \boldsymbol{z}_{\xi_k}] \mid \mathcal{F}_{k-1}\} = \nabla F[P_C(\boldsymbol{\theta}_{k-1})]. \tag{2.646}$$

$$\min_{\boldsymbol{\theta}} \ F(\boldsymbol{\theta}) + \rho q(\boldsymbol{\theta}), \tag{A.647}$$

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \alpha_k \nabla f(\boldsymbol{\theta}_k, \boldsymbol{z}_{\xi_k}), \tag{648a}$$

$$\nabla F(\boldsymbol{\theta}) + \rho_k[\boldsymbol{\theta} - P_C(\boldsymbol{\theta}_{k-1})] = 0, \tag{649}$$

$$\nabla F(\boldsymbol{\theta})^T(\boldsymbol{\theta} - \boldsymbol{\theta}_*) \geq F(\boldsymbol{\theta}) - F(\boldsymbol{\theta}_*) + \frac{\mu}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_*\|^2. \tag{6.650}$$

$$\delta_k \leq \frac{2m(1 + \frac{\mu}{\rho_1})}{\mu\rho_1}k^{-\gamma} + (1 + \frac{\mu}{\rho_1})^{-\phi_\gamma(k)}(\delta_0 + A) \tag{A.651}$$

$$y_k \leq \exp(\sum_{i=1}^{+\infty} a_i) \cdot \sum_{i=1}^{+\infty} b_i + \exp(\sum_{i=1}^{+\infty} a_i) \cdot y_0. \tag{652a}$$

$$\|P_C(\boldsymbol{\theta}_{i_k-1}) - \boldsymbol{\theta}_*\|^2 \to Y. \tag{653}$$

$$\mathbb{P}(\lim_{k\to\infty}\|P_C(\boldsymbol{\theta}_{i_k-1}) - \boldsymbol{\theta}_*\|^2 > 0) \ \leq \ \mathbb{P}\big(\lim_{k\to\infty} F[P_C(\boldsymbol{\theta}_{i_k-1})] - F(\boldsymbol{\theta}_*) > 0\big) = 0. \tag{3.654}$$

$$\delta_n \triangleq \frac{1}{a_n}(a_{n-1}/b_{n-1} - a_n/b_n). \tag{A.655}$$

$$y_n \leq 2\frac{a_1(1+b_1)}{b_1}n^{-\alpha+\beta} + \exp(-\log(1+b_1)\phi_\beta(n))[y_0 + (1+b_1)^{n_0}A], \tag{656a}$$

$$y_i \sim \text{Binom}(n_i, p_i) \tag{657}$$

$$\mathbb{P}\left(p_i > p_{0i}|\mathbf{y}\right) > 85\% \tag{7.658}$$

$$\begin{aligned}\Delta_\theta(v,x) &:= g_{\theta+v}(x) - g_\theta(x)\\ \psi(\theta,v,q) &:= \log \mathbb{E}_\theta\left[e^{q\Delta_\theta(v,X)}\right]\end{aligned} \tag{A.659}$$

$$f(\theta_0+v) \leq f(\theta_0)^{1-\frac{1}{q}} \exp\left[\frac{\psi(\theta_0,v,q)}{q} - \psi(\theta_0,v,1)\right] \tag{660a}$$

$$f(\theta_0+v) \leq \inf_{q\in[1,\infty]} U(\theta_0,v,q,f(\theta_0)) \tag{661}$$

$$\sup_{v\in H} f(\theta_0+v) \leq \inf_{q\in[1,\infty]} \sup_{v\in H} U(\theta_0,v,q,f(\theta_0)) \tag{4.662}$$

$$\{\mathcal{N}(\theta,1) : \theta\in\Theta\} \tag{A.663}$$

$$U(\theta_0,v,q,a) = a^{1-\frac{1}{q}} \exp\left(\frac{(q-1)v^2}{2}\right) \tag{664a}$$

$$\hat{u} := \text{Beta}^{-1}(1-\delta, R+1, N-R) \tag{665}$$

$$\mathbb{P}_{\theta_0}\left(f(\theta_0+v) \leq U(\theta_0,v,q,\hat{u}_0)\right) \geq \mathbb{P}\left(f(\theta_0) \leq \hat{u}_0\right) \geq 1-\delta \tag{1.666}$$

$$\forall\theta\in\Theta,\ \mathbb{P}\left(f(\theta) \leq \hat{u}(\theta)\right) \geq 1-\delta \text{ and } \mathbb{P}\left(f(\theta) \geq \hat{\ell}(\theta)\right) \geq 1-\delta \tag{A.667}$$

$$\forall\theta\in\Theta,\ \mathbb{E}\left[f_{\hat{\lambda}^*}(\theta)\right] \leq \alpha \tag{668a}$$

$$f(\theta_0+v) \leq a + (b-a)\left(\frac{f(\theta_0)-a}{b-a}\right)^{1-\frac{1}{q}} \exp\left[\frac{\psi(\theta_0,v,q)}{q} - \psi(\theta_0,v,1)\right] \tag{669}$$

$$1 - f(\theta_0+v) \leq (1-f(\theta_0))^{1-\frac{1}{q}} \exp\left[\frac{\psi(\theta_0,v,q)}{q} - \psi(\theta_0,v,1)\right] \tag{5.670}$$

$$\Delta(v,x) = g_{\theta_0+v}(x) - g_{\theta_0}(x) = T(x)^\top v \tag{A.671}$$

$$p_\theta(y) = \exp\left[g_\theta(y) - A(\theta)\right] \tag{672a}$$

$$\Delta(v,y) = g_{\theta_0+v}(y) - g_{\theta_0}(y) = y^\top X v = T_X(y)^\top v \tag{673}$$

$$D_\alpha(Q\|P) = \frac{1}{\alpha-1}\log \mathbb{E}_P\left[\left(\frac{dQ}{dP}\right)^\alpha\right] \tag{2.674}$$

$$U(\theta_0, v, q, a) = a^{1-\frac{1}{q}} \exp\left\{\left(1 - \frac{1}{q}\right) D_q(P_{\theta_0+v} \| P_{\theta_0})\right\} \tag{A.675}$$

$$f(\theta_0 + v) \leq \inf_{q \geq 1} U(\theta_0, v, q, f(\theta_0)) = \inf_{q \geq 1} \left\{ f(\theta_0)^{1-\frac{1}{q}} \exp\left(\frac{(q-1)v^2}{2}\right)\right\} \tag{676a}$$

$$f(\theta_0 + v) \leq f(\theta_0) + f'(\theta_0)v + \frac{v^2}{2} \tag{677}$$

$$\sup_{v \in H_i - \theta_i} f_{b(i)}(\theta_i + v) \leq \inf_{q \geq 1} \sup_{v \in H_i - \theta_i} U(\theta_i, v, q, f_{b(i)}(\theta_i)) \tag{6.678}$$

$$C^e_{1-\alpha} := \{e(\theta) : \theta \in C_{1-\alpha}\} \tag{A.679}$$

$$\begin{aligned} y_i &= g(x_i) + \epsilon_i \\ \epsilon_i &\sim \mathcal{N}(0, 1) \end{aligned} \tag{680a}$$

$$d_{L^2_X}(a, b) := \left(\sum_{i=1}^{n} [a(x_i) - b(x_i)]^2\right)^{\frac{1}{2}} \tag{681}$$

$$\mathbb{P}_{\theta_0}\left(f(\theta_0) \leq \hat{u}\right) \geq 1 - \delta \tag{3.682}$$

$$\hat{u}(\cdot) := \sup_{v \in H - \theta_0} U(\theta_0, v, q, \hat{u}_0) \tag{A.683}$$

$$\psi(\theta_0, v, 1) \equiv A(\theta_0 + v) - A(\theta_0) \tag{684a}$$

$$H_0 : \theta \leq 0 \quad H_1 : \theta > 0 \tag{685}$$

$$\left(-\infty, \hat{\theta}(X) + \lambda u(X)\right) \tag{7.686}$$

$$\mu(x) = h(x) + c\,, \tag{A.687}$$

$$h(x) = ae^{bx}\,, \tag{688a}$$

$$h(x) = \frac{ae^{bx}}{1 + \frac{a\gamma}{b}(e^{bx} - 1)}\,, \tag{689}$$

$$h(x) = a_1 e^{-b_1 x} + ae^{bx}\,. \tag{4.690}$$

$$\mu(x) = h(x, 1) + h(x, 2) + \cdots + h(x, n)\,, \tag{A.691}$$

$$f(x) = \sum_{j=1}^{n} \pi_j g_j(x)\,, \tag{692a}$$

$$S_Y(t) = \mathbb{P}\left(Y > t\right) = S_{\mathbf{T}}(t, \cdots, t)\,, \tag{693}$$

$$S_Y(t) = \sum_{j=1}^{n} \pi_j S(t \,|\, j)\,. \tag{1.694}$$

$$f_Y(t) = -\frac{d}{dt} S_Y(t) = \sum_{j=1}^{n} f(t,j) = \sum_{j=1}^{n} \pi_j f(t \,|\, j)\,, \tag{A.695}$$

$$h_Y(t) = \frac{f_Y(t)}{S_Y(t)} = \sum_{j=1}^{n} h(t,j)\,, \tag{696a}$$

$$f(x) = \pi g_1(x) + (1-\pi) g_2(x), \tag{697}$$

$$\pi = \int_0^\infty c \exp\left\{-cx - H(x)\right\} dx\,, \tag{5.698}$$

$$x^* = \max\{\max\{x : h(x) = c\}, 0\}\,. \tag{A.699}$$

$$p(x) = c/(h(x) + c)\,. \tag{700a}$$

$$h(x \,|\, 1) = \frac{\exp\left\{-cx - H(x)\right\}}{\int_x^\infty \exp\left\{-cy - H(x)\right\} dy}\,. \tag{701}$$

$$h(x) + c = \frac{dh(x)/dx}{h(x)}\,. \tag{2.702}$$

$$M_{GM} = \frac{1}{b} \ln \frac{b-c}{a}\,, \tag{A.703}$$

$$M_{\Gamma GM} = \frac{1}{b} \ln \left(\frac{b}{a} \cdot \frac{b - a\gamma - c\left(1 - \frac{a\gamma}{b}\right)}{b + a\gamma c} \right)\,. \tag{704a}$$

$$M_{BM} = \frac{1}{b} \ln \frac{b-c}{a(1+ck)}\,, \tag{705}$$

$$\pi = \frac{c}{a} \left(\frac{a}{b}\right)^{\frac{c}{b}} e^{\frac{a}{b}} \Gamma\left(-\frac{c}{b}, \frac{a}{b}\right) \tag{6.706}$$

$$S(x) = \exp\left\{-\frac{a}{b}\left(e^{bx} - 1\right) - cx\right\}\,. \tag{A.707}$$

$$f(x) = c \exp\left\{-\frac{a}{b}\left(e^{bx} - 1\right) - cx\right\} + ae^{bx} \exp\left\{-\frac{a}{b}\left(e^{bx} - 1\right) - cx\right\}. \tag{708a}$$

$$g_1(x) = \frac{c}{\pi} \exp\left\{-\frac{a}{b}\left(e^{bx} - 1\right) - cx\right\}, \quad g_2(x) = \frac{ae^{-(c-b)x}}{1-\pi} \exp\left\{-\frac{a}{b}\left(e^{bx} - 1\right)\right\} \tag{709}$$

$$S(x) = S_1(x)\, S_2(x) = \pi G_1(x) + (1-\pi) G_2(x) \tag{3.710}$$

$$x^* = \begin{cases} \frac{1}{b} \ln \frac{c}{a} & c \geq a \\ 0 & c < a \end{cases} \tag{A.711}$$

$$S_{\mathbf{T}}(t_1, \cdots, t_n) = \mathbb{P}\left(\bigcap_{j=1}^{n} [T_j > t_j]\right), \tag{712a}$$

$$S_Y(t) = \sum_{j=1}^{n} \mathbb{P}\left(Y > t, J = j\right) = \sum_{j=1}^{n} \mathbb{P}\left(Y > t \,|\, J = j\right) \mathbb{P}\left(J = j\right). \tag{713}$$

$$S(t \,|\, j) = \mathbb{P}\left(Y > t \,|\, J = j\right) = \frac{\mathbb{P}(t < T_j < \min_{i \neq j} T_i)}{\mathbb{P}(T_j < \min_{i \neq j} T_i)}, \tag{7.714}$$

$$\pi_j = \mathbb{P}\left(J = j\right) = \mathbb{P}(T_j < \min_{i \neq j} T_i) \tag{A.715}$$

$$h(t, j) = h_j(t), \quad j = 1, \ldots, n. \tag{716a}$$